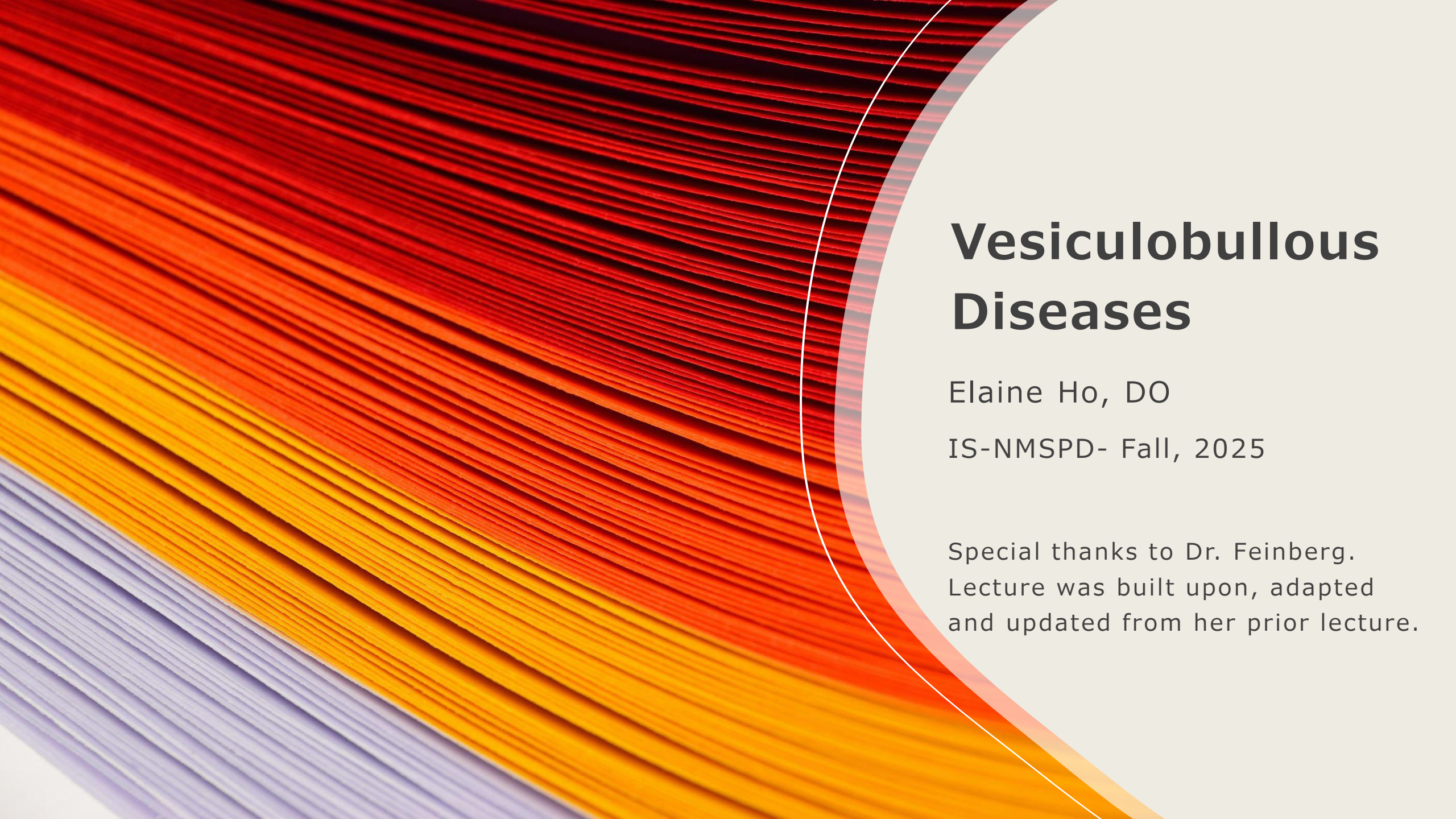




Vesiculobullous Diseases

Elaine Ho, DO

IS-NMSPD- Fall, 2025

Special thanks to Dr. Feinberg.
Lecture was built upon, adapted
and updated from her prior lecture.

Objectives

For each of the following Disorders, know the objectives A-C

- Pemphigus
 - o **Pemphigus vulgaris**
 - o **Pemphigus foliaceus**
 - o Paraneoplastic pemphigus
 - o Hailey-Hailey disease
- **Bullous pemphigoid**
- Mucous membrane pemphigoid
- Epidermolysis
- Bullosa acquisita
- **Dermatitis herpetiformis**
- Linear IgA bullous dermatosis
- Epidermolysis bullosa
- **Bullous Drug eruptions**
- Various infectious vesiculobullous diseases

Objectives

- A. Discuss approach to cutaneous blisters
- B. List causes/triggers
- C. Describe clinical presentation, workup, and management

Definitions

Definitions:

- Vesicle= elevated, clear fluid filled lesion $\leq 1\text{cm}$ in diameter
- Bulla= elevated, clear fluid filled lesion $> 1\text{cm}$ in diameter
- Blistering skin disorders can be characterized based on which layer the skin is pulling away from (which layer the blister is)
 - The more superficial= fragile blisters
 - The deeper= tense blisters
 - *Biopsy* is a helpful way for diagnosis, because you can define the blister by getting a sample and looking at which layer is affected.
- Acantholysis= loss of adhesions between cells within the epidermis
- Spongiosis= edema between epidermal cells, or disassociation of epidermis and dermis

Stratum corneum

Stratum lucidum*

Stratum granulosum

Stratum malphigii
(spinosum)

Stratum germinativum
(basale)

Dermal papillae

Papillary

Reticular

Nerve endings

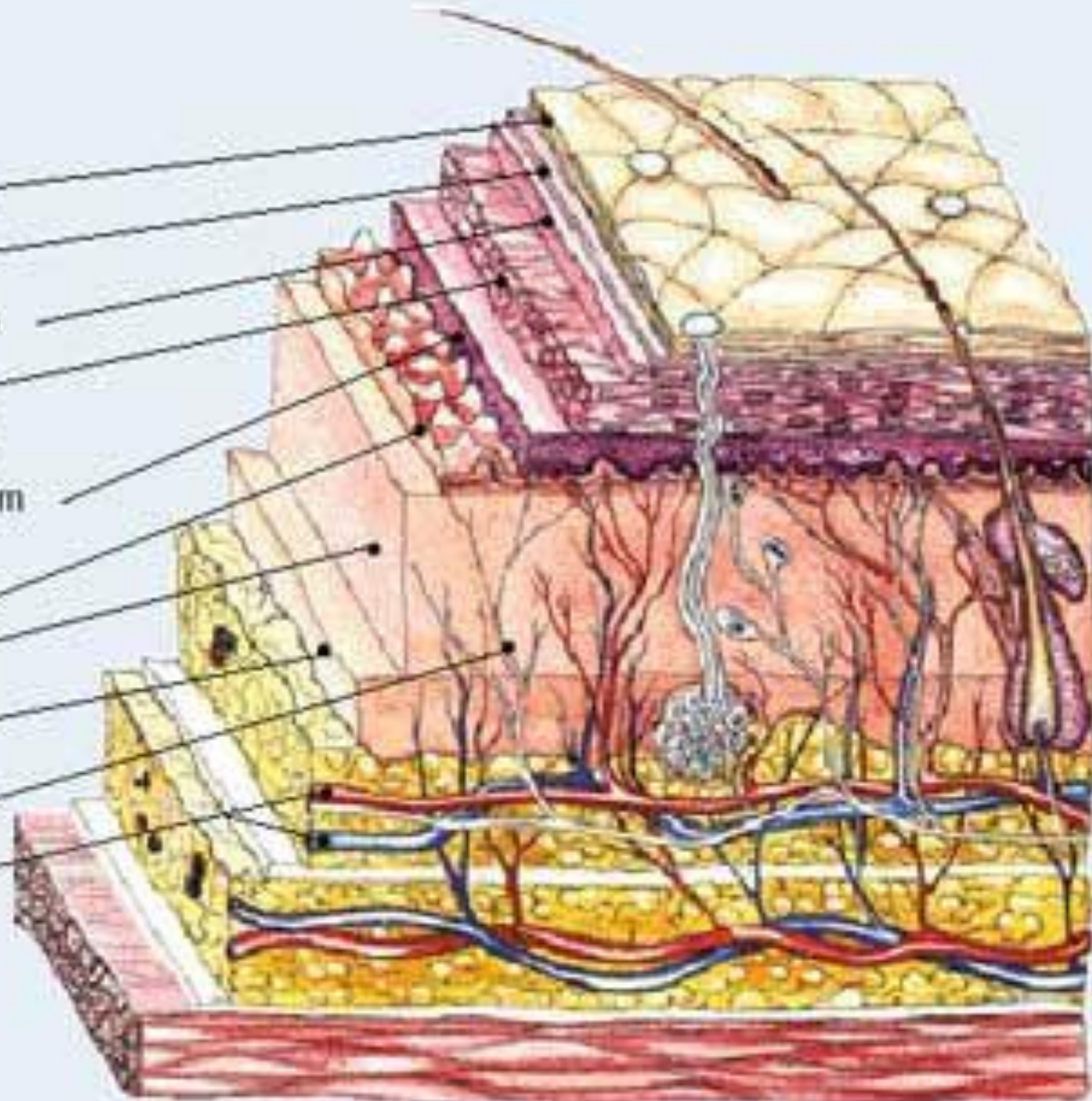
Blood vessels

Epidermis

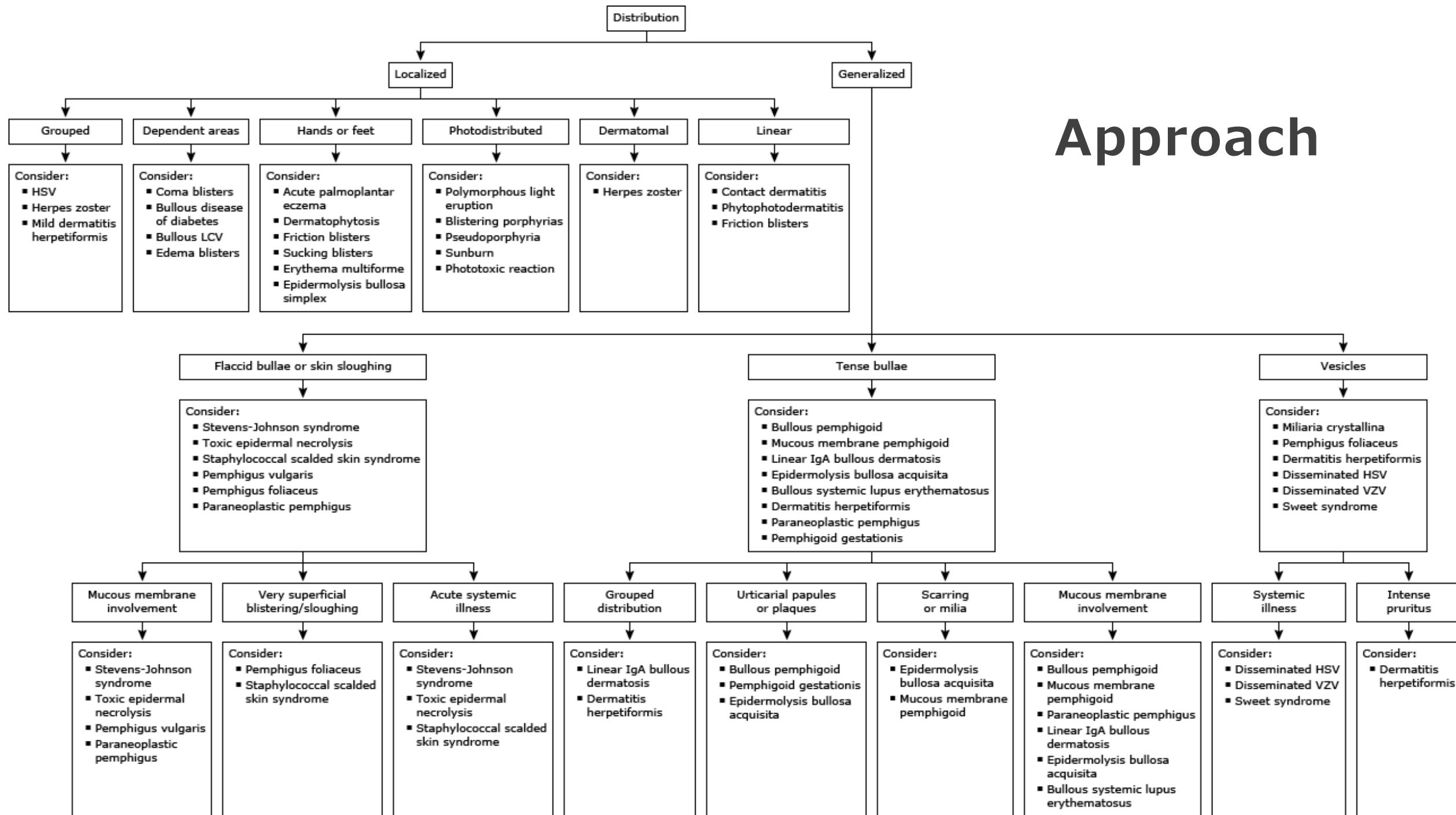
Dermis

Subcutaneous
tissue

Muscle



Approach



History

- Where are the lesions located (generalized, localized, specific sites)?
- Are mucous membranes involved?
- What is the size and configuration of blisters?
- If bullae are present, are they flaccid or tense?
- What is the patient's age?
- New medication?
- Exacerbations/exposures?
- Immunosuppression?

Physical

- **Lesion distribution** — The identification of cutaneous blisters as generalized, localized, or associated with mucous membrane involvement can offer valuable clues for diagnosis
 - **Localized vs generalized**
 - **Flaccid bullae vs tense bullae vs vesicles**

Physical

- **Nikolsky sign**
 - Eliciting skin blistering from gentle mechanical pressure on sign. This may be observed at the edge of existing lesion, in area of normal-appearing skin, or both locations
 - [Nikolsky's sign \(Toxic epidermal necrolysis\) \(youtube.com\)](#)



Physical

- **Asboe-Hansen sign**
 - Lateral spreading of a preexisting intact bulla by applying pressure to it
 - <https://www.youtube.com/shorts/NWlgWWZxOLg>



Workup

- What are the ddx? What can I order to help me navigate the ddx?
- Which layer is the blister occurring?
- Antibodies involved?
- Genetic component?

***Pemphigus
Vulgaris***

Causes / Population

- Causes
 - Mostly idiopathic
 - Sometimes drug-induced
- Population
 - 40-60yo
 - male=female
 - More common in
 - Ashkenazi Jewish
 - Southeast Europe
 - Middle East
 - India

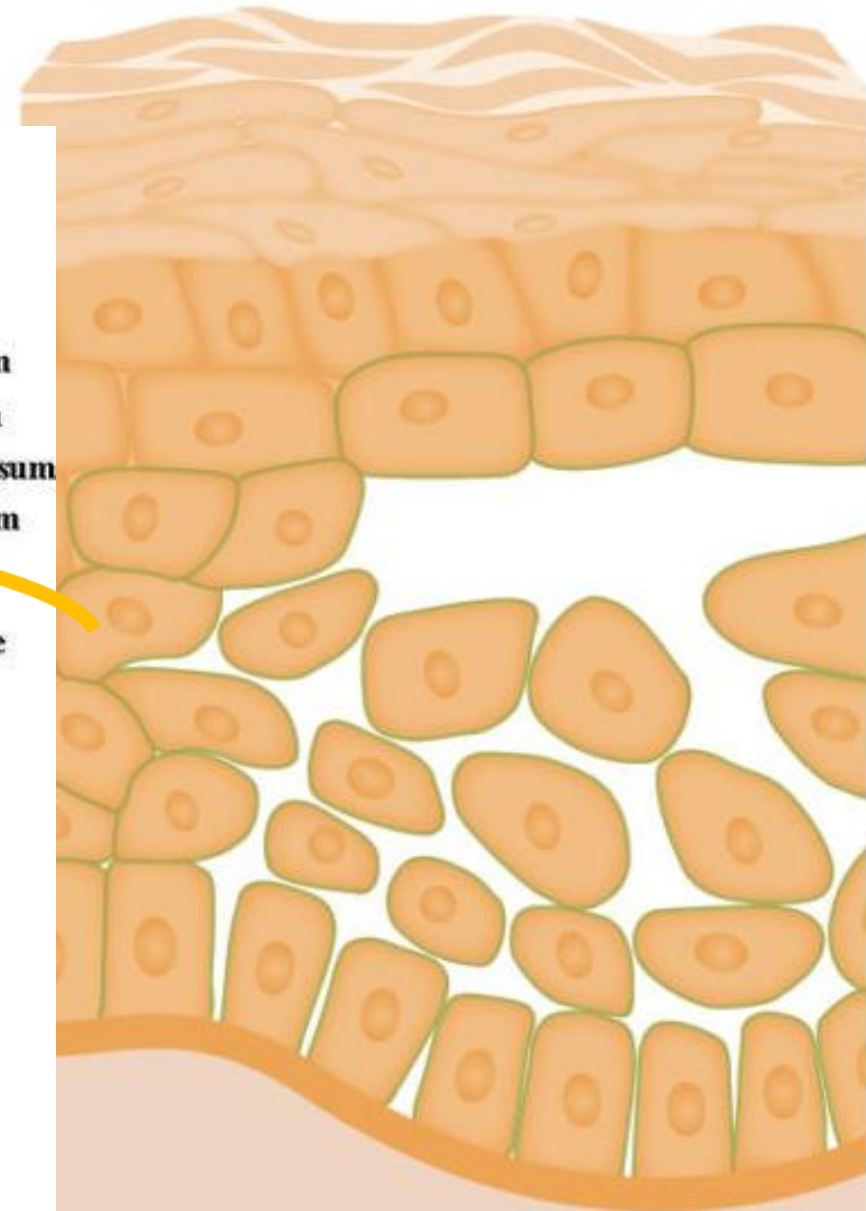
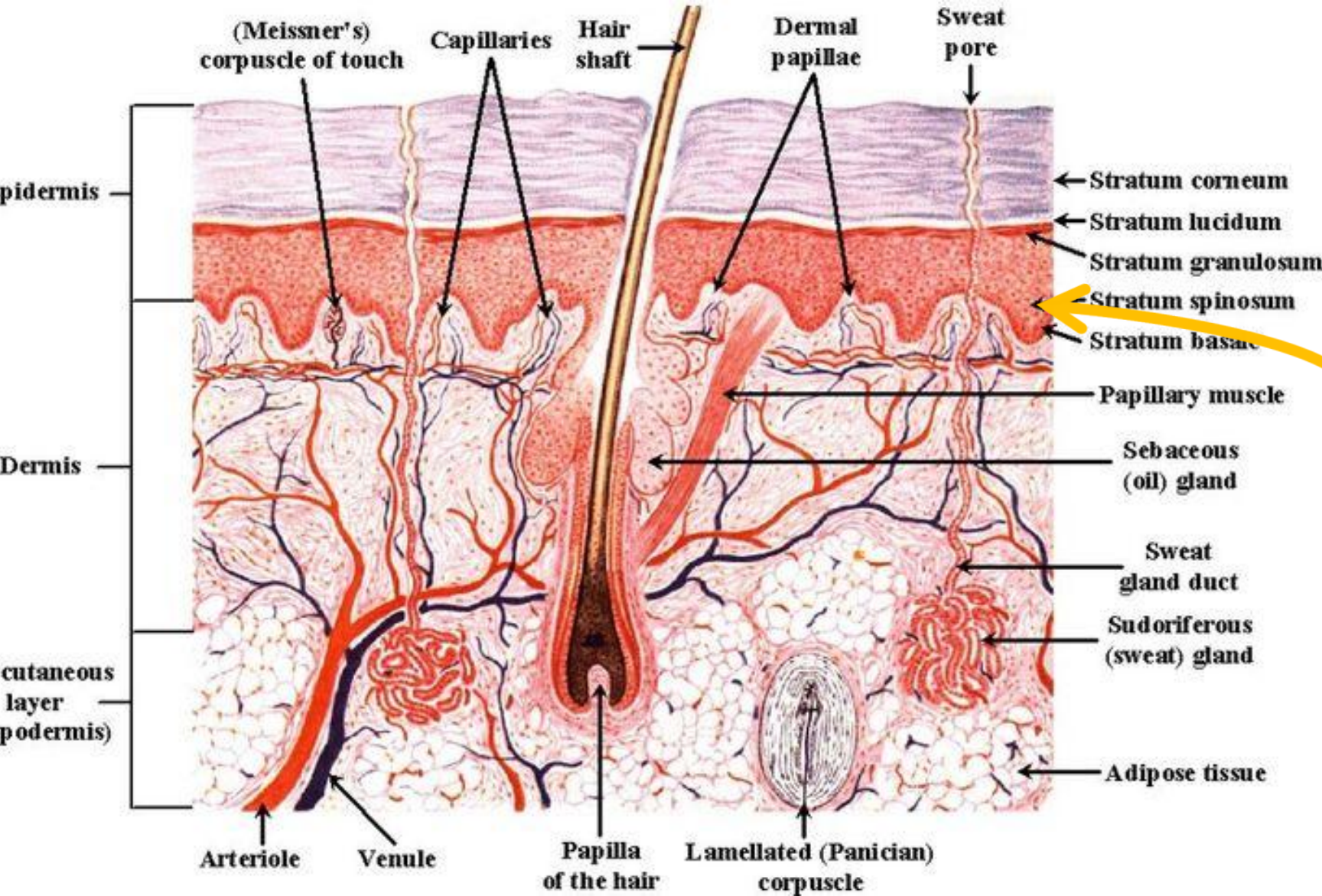
IgG Auto-Antibodies to:

- Desmosomal proteins
 - Desmoglein 3
 - Sometimes Desmoglein 1

Layer that separates

- Spinosum layer of epidermis

B Pemphigus vulgaris





Clinical Presentation

- Erosions > Bullae
 - *Fragile/flaccid bullae unroof → erosion*

Clinical Presentation

- Intact Bullae



Clinical Presentation

- Erosions and unroofed Bullae



Clinical Presentation – clinical variant

- Pemphigus vegetans
 - vegetating plaques composed of excessive granulation tissue and crusting
- Often in groin and axilla



Location

- *Mucosal AND cutaneous involvement*
- Common areas:
oral mucosa (lips, gum, palate)



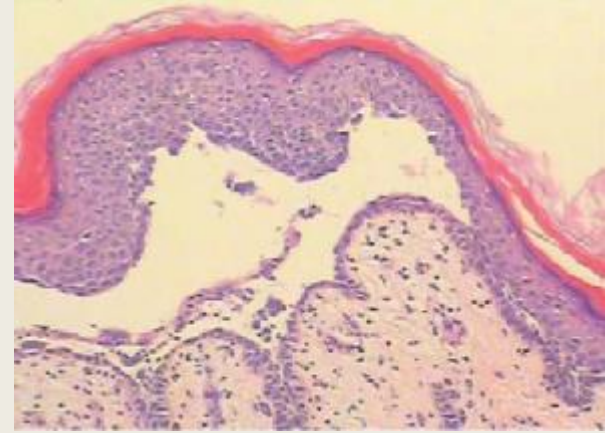
Signs

- + Nikolsky's sign
 - (some do not recommend to do this if patient already has erosive skin lesions)
- + Asboe-Hansen sign

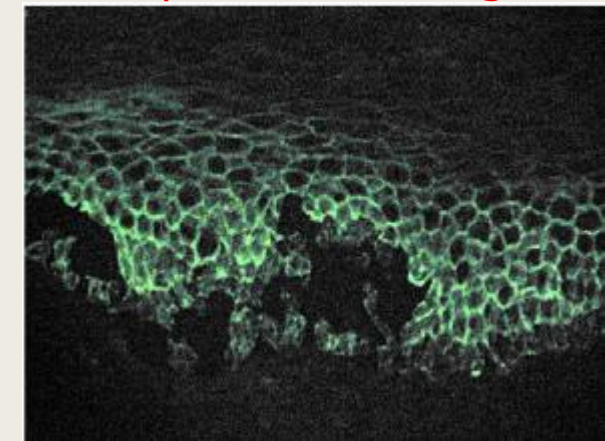
Complications

- ***Severe flares are often fatal if untreated***
 - Rate of skin denudation exceeds rate of skin repair → extensive skin erosions → predispose to secondary infection / protein loss / fluid balance disturbance
 - Mouth pain prevents adequate nutritional intake
 - Skin pain limits movement → increased risk for pneumonia and DVTs
- Unpredictable course
- Even with therapy, disease course waxes and wanes
- Permanent, complete remission is difficult to achieve

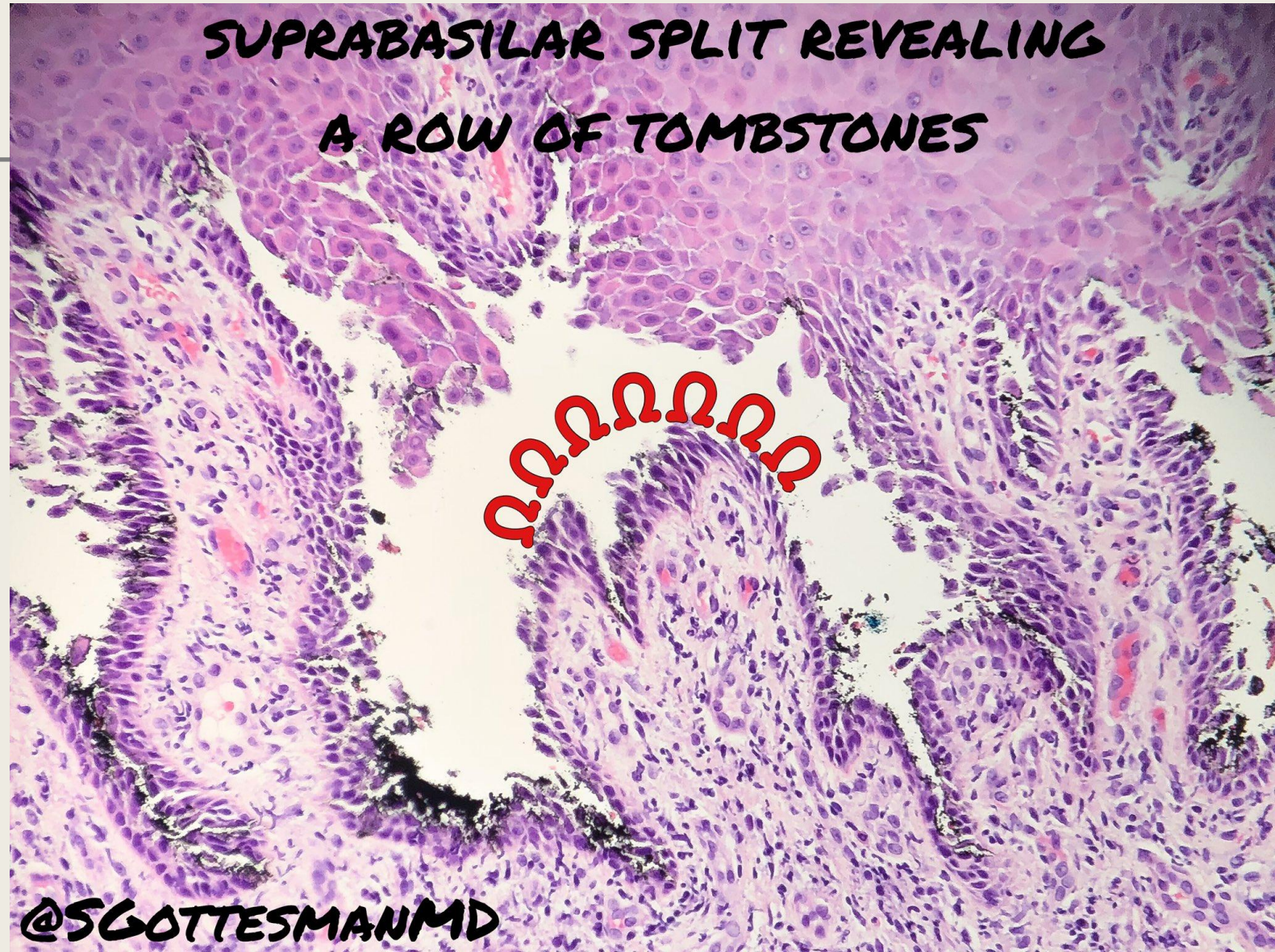
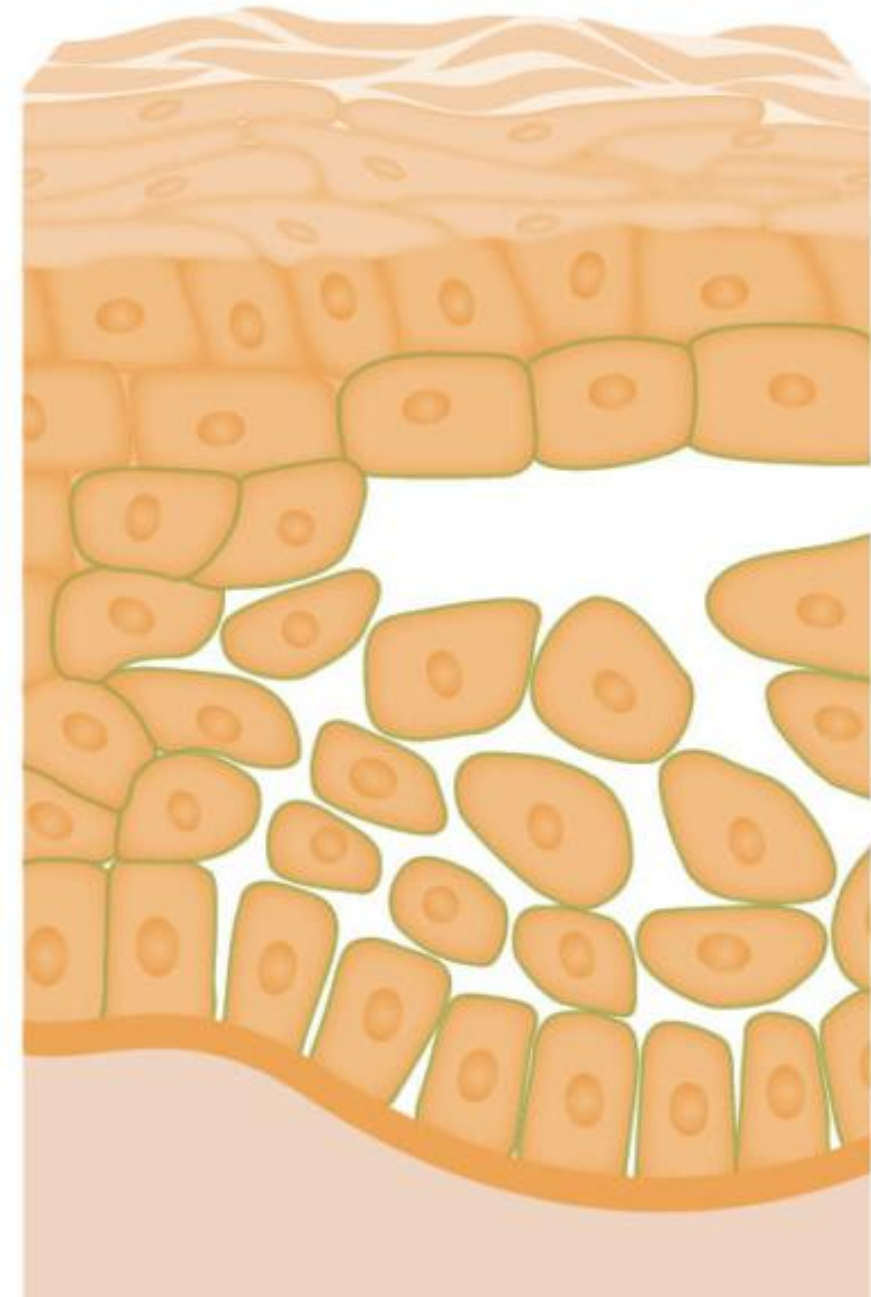
Diagnosis



- Nikolsky sign
- Punch biopsy Biopsy (4mm)
 - Ideal:
 - *Punch* biopsy to get all layers
 - Early lesion
 - 2 samples or Edge of blister/erosion (to get BOTH normal and abnormal skin)
 - Abnormal portion → H&E staining → *suprabasal acantholytic blisters + row of tombstones*
 - Normal portion → direct immunofluorescence showing *intercellular deposition of IgG within epidermis in net-like pattern*
- Lab
 - ELISA
 - *+ IgG abs to desmoglein 1 and desmoglein 3*
 - Indirect immunofluorescence
 - Looks for circulating abs



B Pemphigus vulgaris

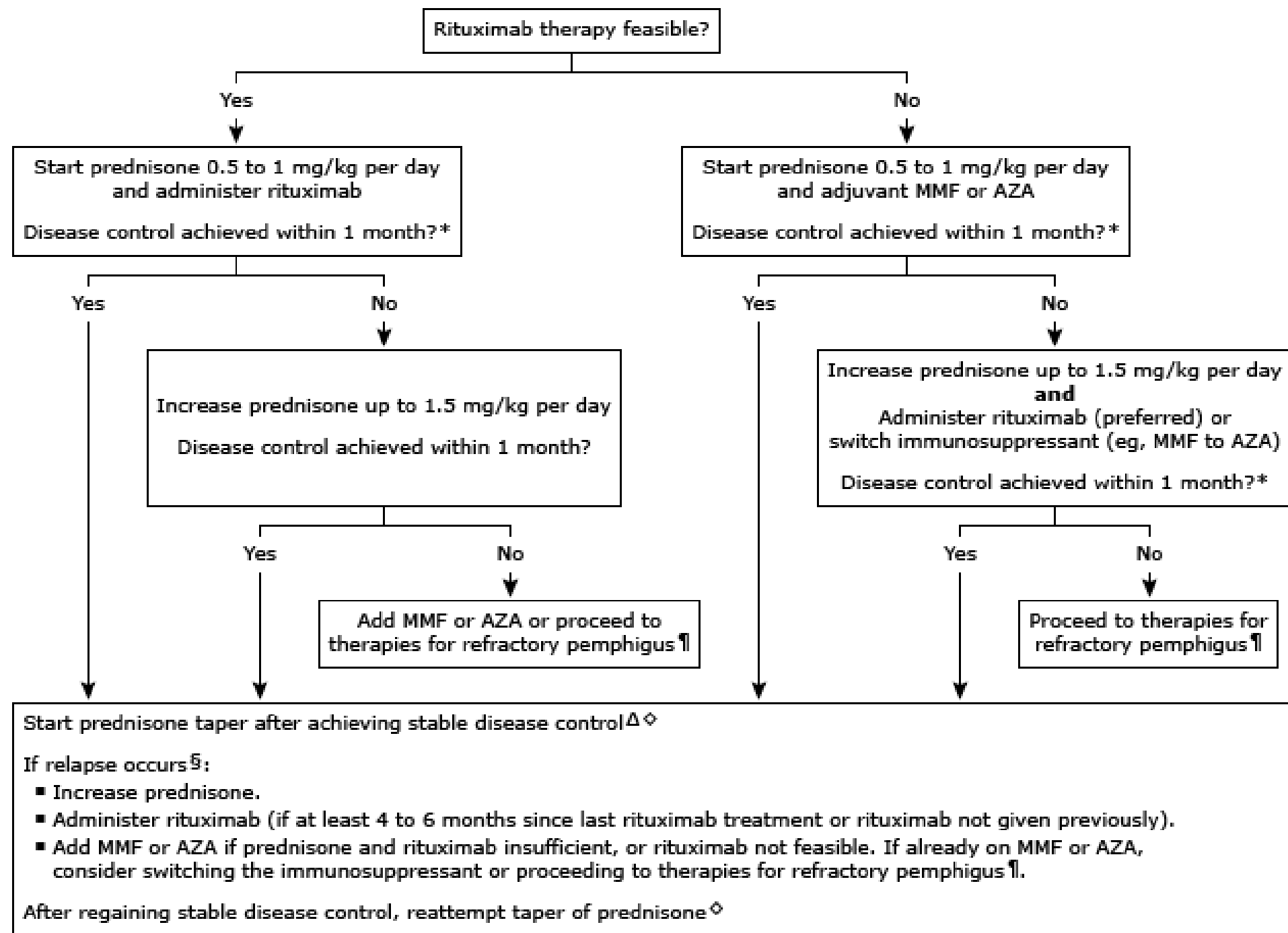


Treatment

- Goal
 - Induce complete remission while minimizing treatment-related adverse effects
 - Long-term remission after discontinuation of therapy is ultimate goal
- Approach
 - Treat at time of disease onset, even if mild
 - *Careful approach to treatment regimen: Note that 5-10% mortality from pemphigus vulgaris result from complications from immunosuppressive therapy*
 - Symptom management with supportive care
 - Taper off slowly after achieving disease control
 - Successful response= no new blisters, current lesions begin to heal, absent Nikolsky sign

Treatment

- Rituximab
+
prednisone



Treatment – Concurrent supportive care

- *Wound care*
 - Blisters/erosions → keep clean, petroleum jelly, nonadhesive wound dressing
 - In persistent lesions topical steroid (ex: clobetasol BID)
 - Monitor for secondary infection (consider HSV or zoster if blisters don't go away)
- Oral discomfort
 - Avoid spicy, sharp, abrasive, hot foods
 - Viscous lidocaine
 - See dentist, keep good oral hygiene
 - Oral topical steroid (ex: triamcinolone dental paste or gel)
 - If get candida, tx with oral antifungal (ex: nystatin, clotrimazole)
- Patient support
 - [International Pemphigus and Pemphigoid Foundation](#)

Prognosis

- Most patients enter partial remission within 2-5 years



Pemphigus Foliaceus

Causes / Population

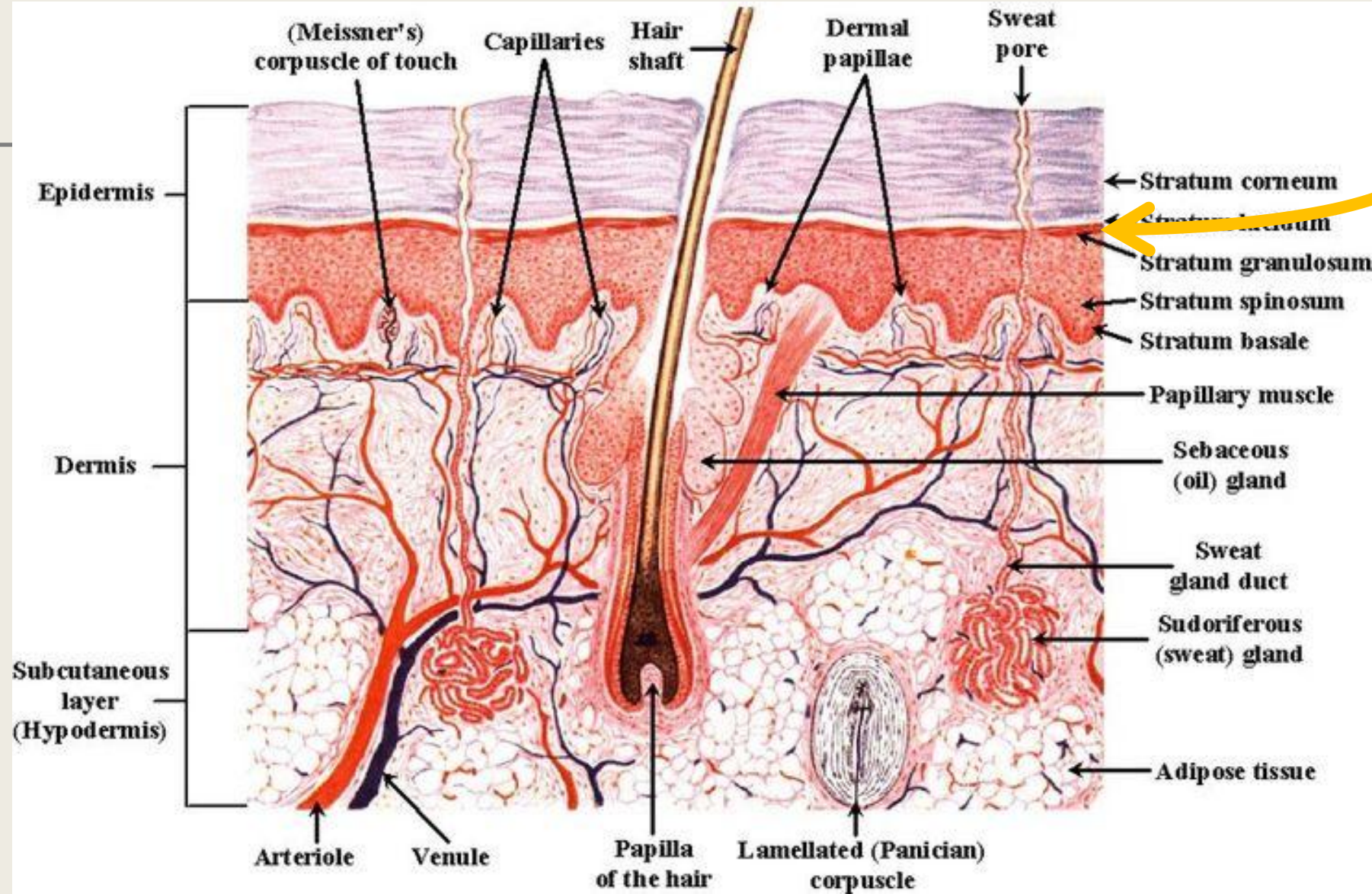
- Causes
 - Mostly idiopathic
 - In endemic areas, black fly/insects serve as vector
 - Drug induced. Ex: captopril, penicillins, cephalosporins
- Population
 - 40-60yo
 - male=female
 - More common in
 - Tunisia
 - Brasil

IgG Auto-Antibodies to:

- Desmosomal proteins
 - Desmoglein 1

Layer that separates

- Granulosum layer of epidermis





Clinical Presentation

- Small, scattered superficial blisters
→ rapidly evolve into scaly, crusted erosions

Clinical Presentation

- Also affects superficial layer, so bulla rupture easily and you may only see erosions



Clinical Presentation

- Progression



Location

- *cutaneous involvement (mucous membranes typically spared)*
- Common areas: scalp, face, trunk (*starts on scalp and face, then later to trunk*)

Signs

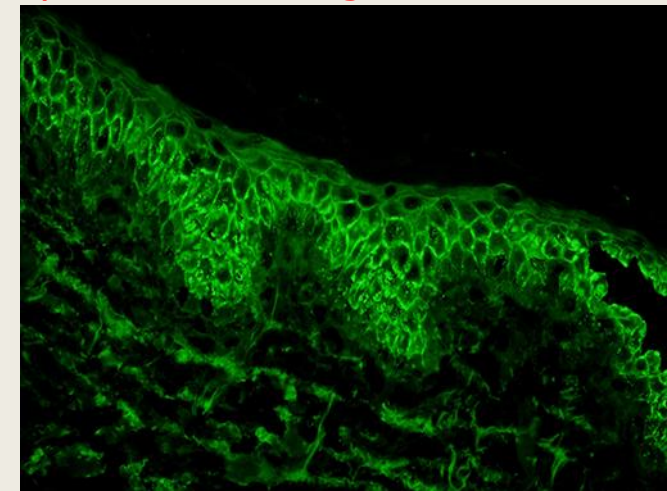
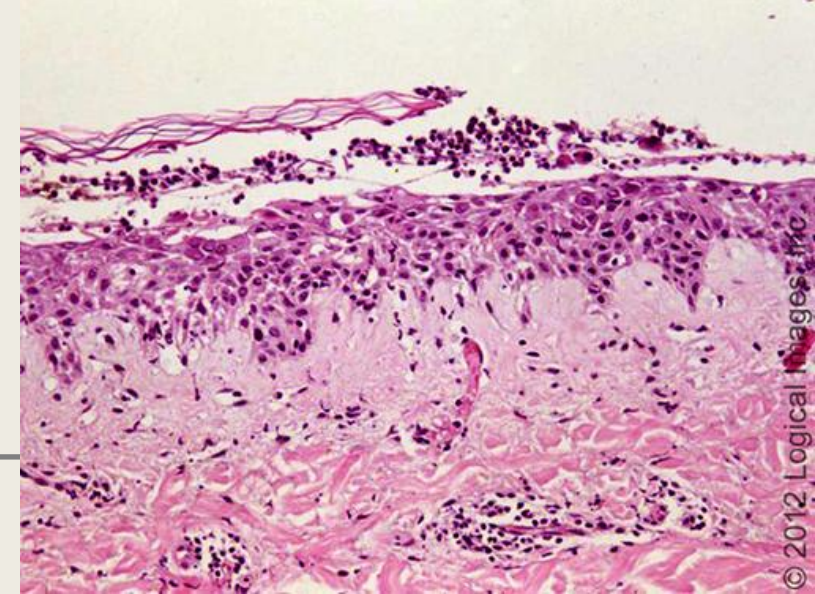
- + Nikolsky's sign
 - (some do not recommend to do this if patient already has erosive skin lesions)

Complications

- Similar to pemphigus vulgaris

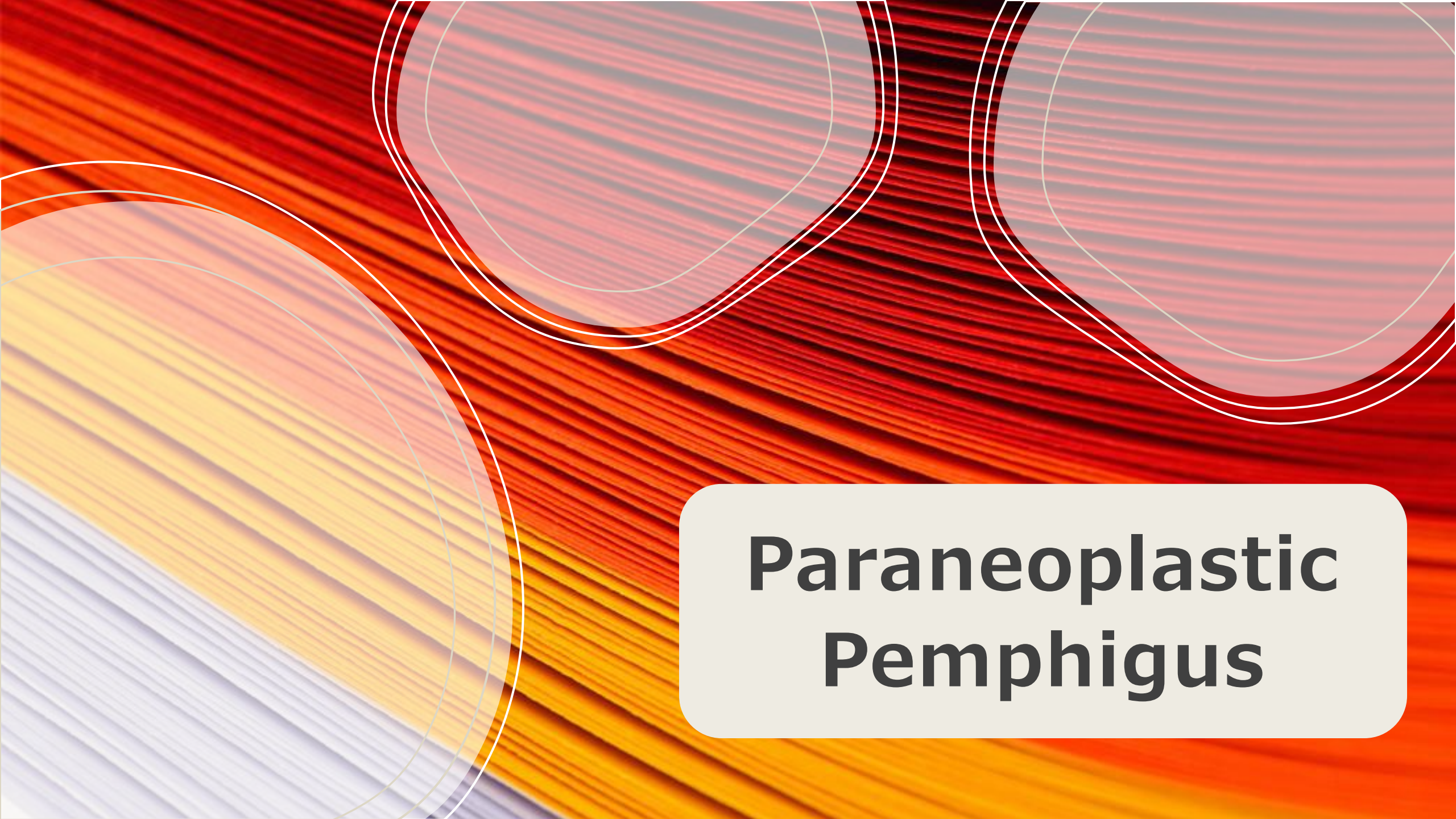
Diagnosis

- Nikolsky sign
- Punch biopsy Biopsy (4mm)
 - Ideal:
 - *Punch* biopsy to get all layers
 - Early lesion
 - 2 samples or Edge of blister/erosion (to get BOTH normal and abnormal skin)
 - Abnormal portion → H&E staining → granular or sub stratum corneal acantholytic blisters
 - Normal portion → direct immunofluorescence showing intercellular deposition of IgG within upper epidermis in net-like pattern
- Lab
 - ELISA
 - + IgG abs to desmoglein 1
 - Indirect immunofluorescence
 - Looks for circulating abs



Treatment

- Same as pemphigus vulgaris



Paraneoplastic Pemphigus

Causes / Population

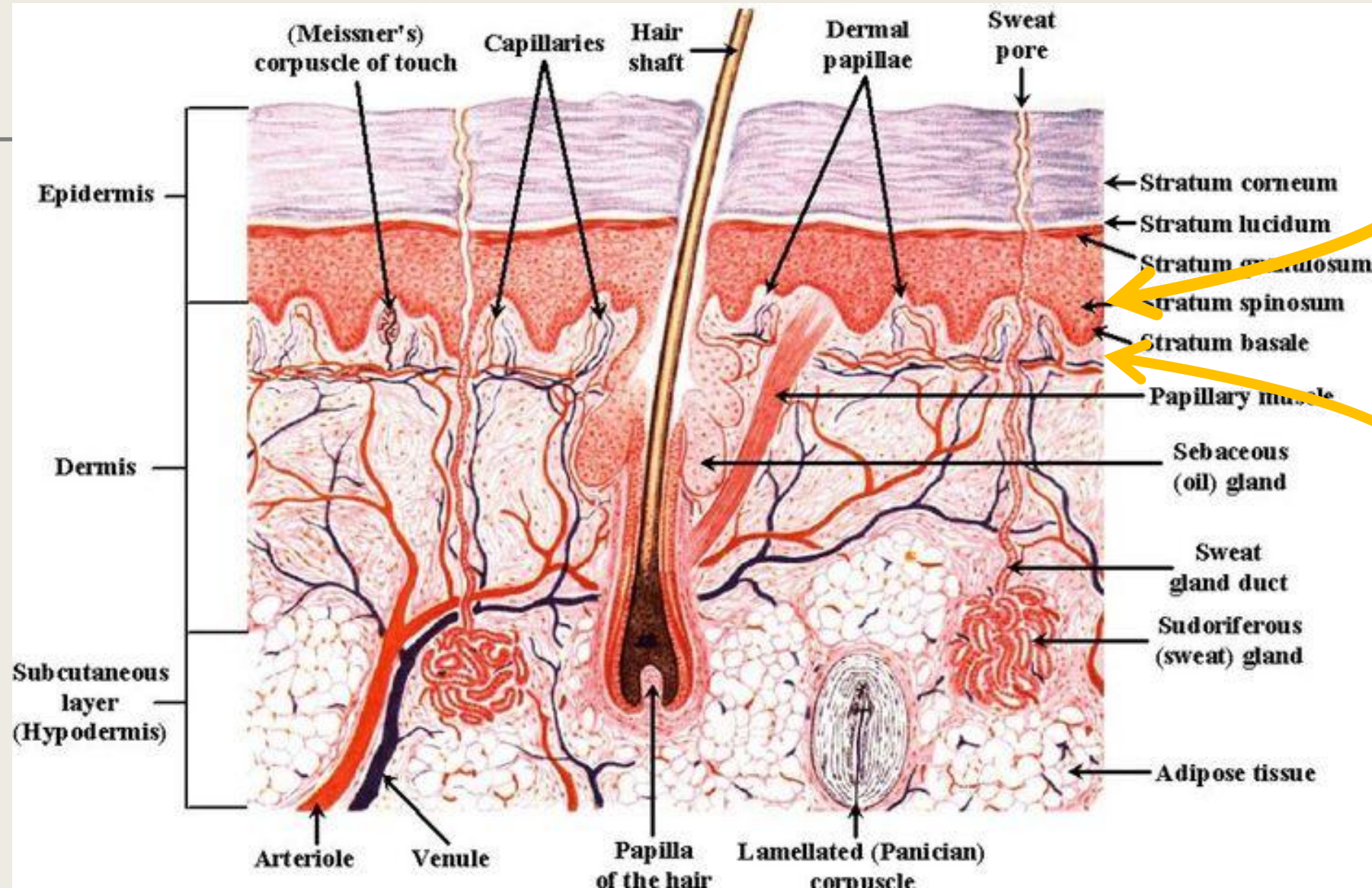
- Cause
 - *Induced by lymphoproliferative disorders* (most commonly non-Hodgkin lymphoma and chronic lymphocytic leukemia), Castleman's disease
- Population
 - 45-70yo
 - male=female
 - Rare, frequency decreasing (due to increased use of anti-CD20 monoclonal antibody (ex: rituximab) for tx of non-Hodgkin lymphoma and CLL)

IgG Auto-Antibodies to:

- *Desmoplakins*
 - *Periplakin and envoplakin*
 - (most specific lab finding)

Layer that separates

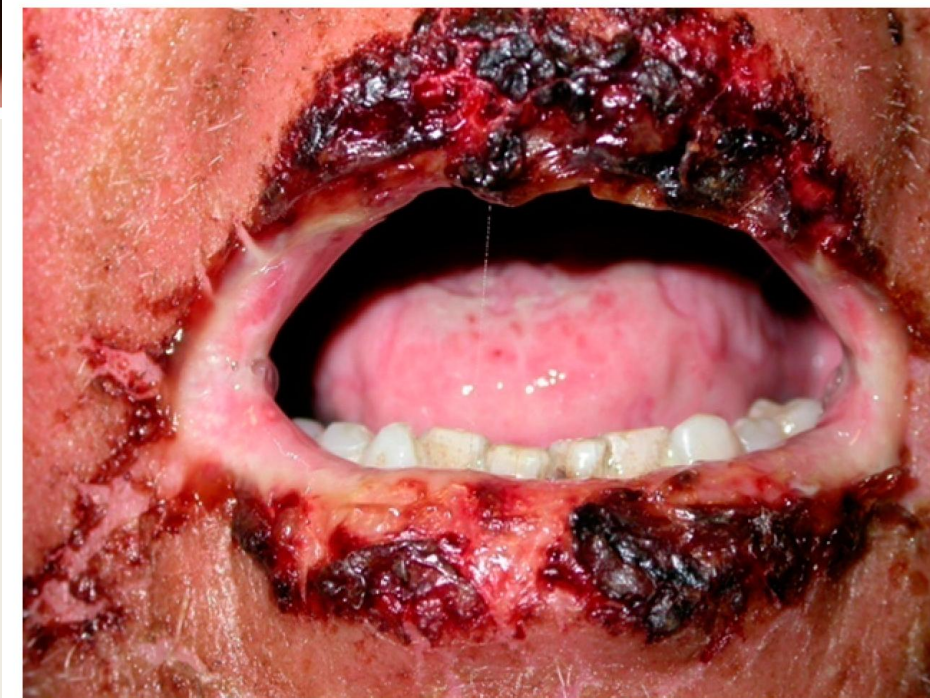
- *Varies...*
- Suprabasal
- subepidermal





Clinical Presentation

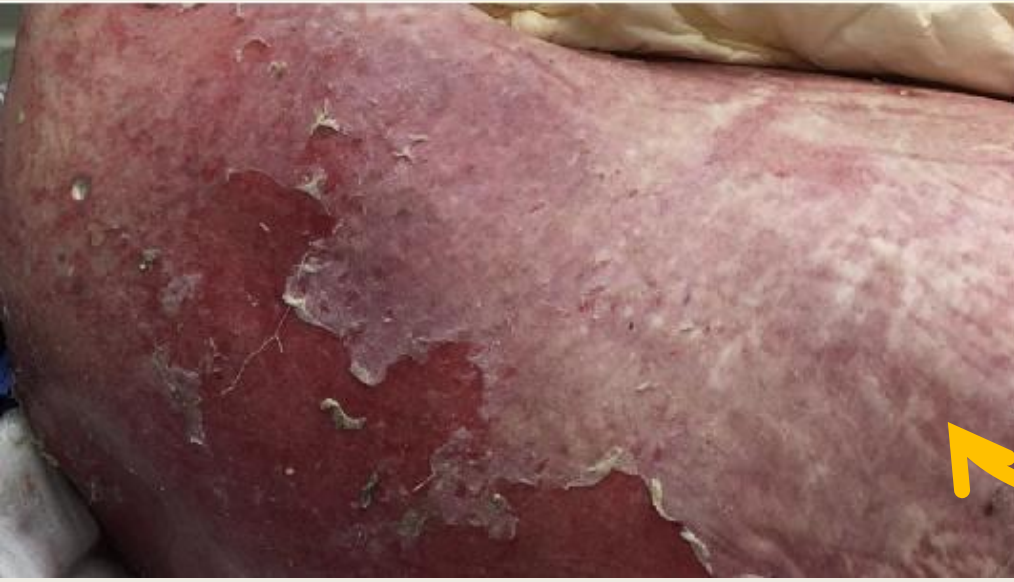
- Hemorrhagic crusting of **lips** with refractory **stomatitis**



Clinical Presentation

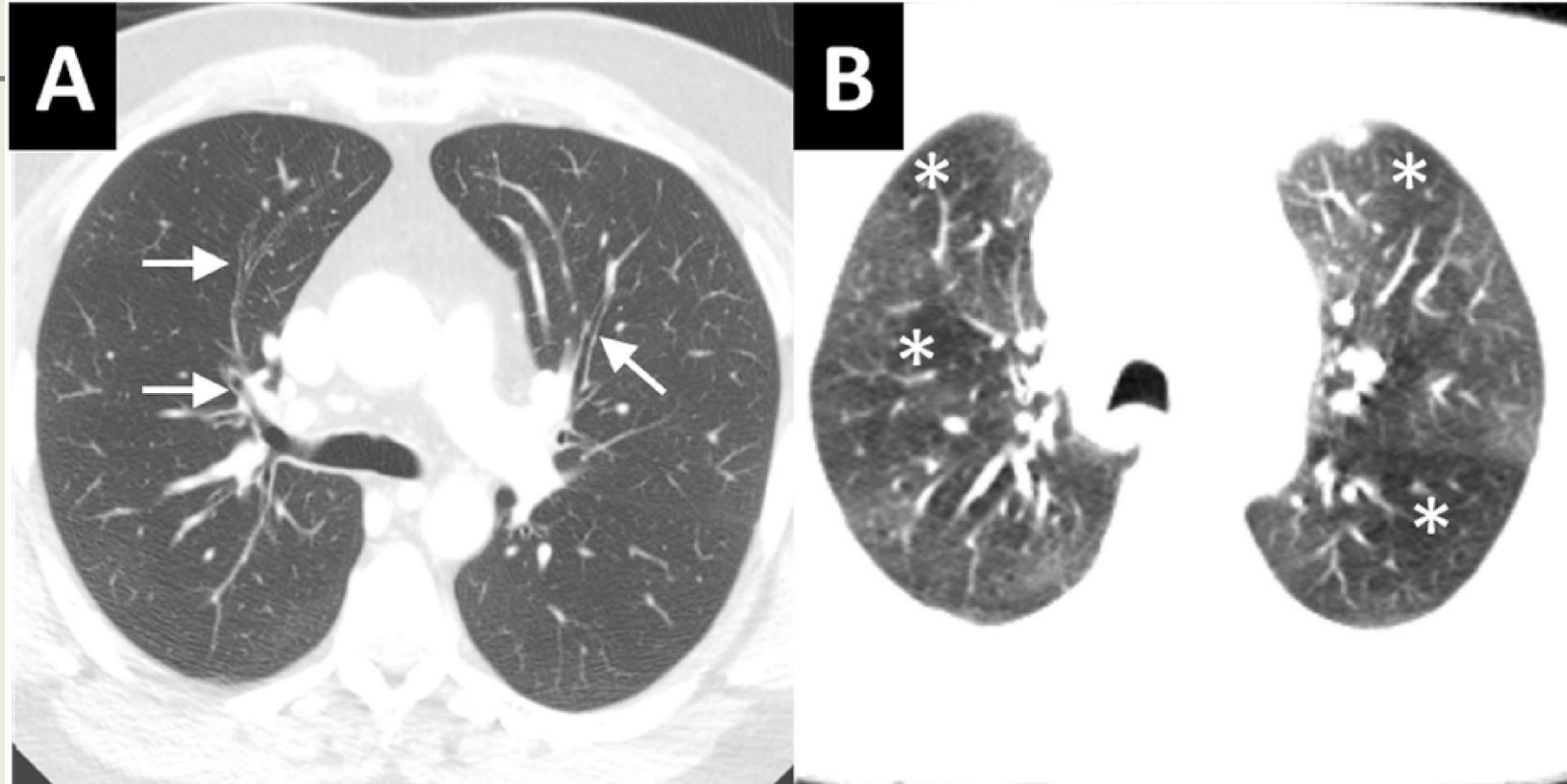


- Variable skin lesions:
 - both flaccid and tense bullae
 - Inflammatory papules or plaques
 - Targetoid lesions
 - Extensive cutaneous desquamation



Clinical Presentation

- *Bronchiolitis obliterans* – occurs late in course of disease → heralded by shortness of breath that is out of proportion to chest x ray (which shows normal or hyperinflated lungs/maybe some bronchial wall thickening)



From Computed Tomography. A, Inspiratory phase. B, Expiratory phase. Mild, diffuse bronchial wall thickening (arrows) and patchy areas of air-trapping (*) were consistent with bronchiolitis obliterans

Location

- Mucosal (esp mouth) AND cutaneous: usually mucosal occur first, then cutaneous lesions develop
- Systemic features as well: pulmonary disease

Complications

- *Often fatal (90% mortality rate within 1 month to 2 years after dx)*
 - Usually succumb to sepsis, respiratory failure, or underlying malignancy
 - Pulmonary disease often fails to improve despite successful tx of malignancy and resolution of mucocutaneous disease

Diagnosis

- Combined assessment of clinical and lab findings
- ****painful erosive mucositis + polymorphous skin eruption + in setting of neoplastic disease*
- **Punch biopsy Biopsy (4mm)**
 - Ideal:
 - *Punch* biopsy to get all layers
 - Early lesion
 - 2 samples or Edge of blister/erosion (to get BOTH normal and abnormal skin)
 - Abnormal portion → H&E staining → *varying acantholysis, dyskeratosis, lichenoid infiltrate* (depending on skin presentation)
 - Normal portion → *direct immunofluorescence IgG around basement membrane* (varies)

Treatment

- Goal: treat malignancy, suppress disease manifestations, manage symptoms
- **Treat underlying neoplasm**
- *Due to rarity of paraneoplastic pemphigus, no randomized therapeutic trials have been performed... data limited to case reports
- First line: **steroid** (ex: *prednisone 1mg/kg*)+ **rituximab** 375mg/m² weekly x4 weeks as soon as feasible → repeat course q3-6 months → taper prednisone slowly as tolerated

Treatment

- *Supportive treatment*
 - Lidocaine mouthwash for oral pain
 - Narcotic pain meds
 - Vaseline and gauze for cutaneous erosions
 - For toxic epidermal necrolysis-like presentation
→ admit to *burn unit* for wound care
- *Consider *Pneumocystis pneumonia* prophylaxis (due to extensive immunosuppressive therapy)*



Hailey-Hailey Disease

(aka benign familial pemphigus)

Causes / Population

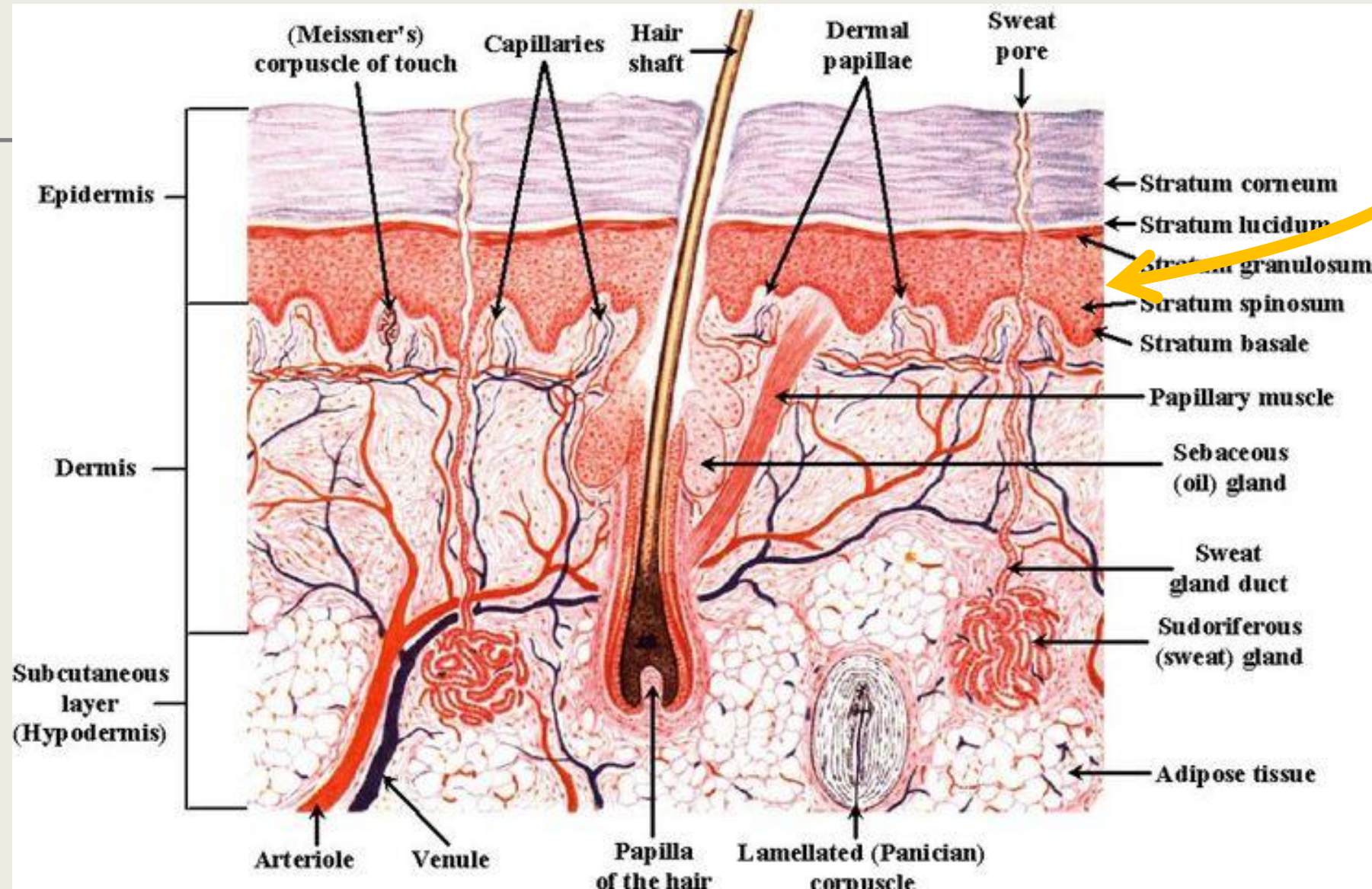
- Cause
 - *Familial*
- Population
 - Unknown prevalence (many lack accurate dx or don't seek treatment)
 - Onset varies within families

IgG Auto-Antibodies to:

- *None*... not autoimmune
- It is autosomal dominant disorder → **ATP2C1 gene defect** → loss of function of calcium magnesium pump → affects Ca^{2+} -dependent signaling pathway in regulating keratinocyte adhesion and differentiation

Layer that separates

- *Suprabasal*





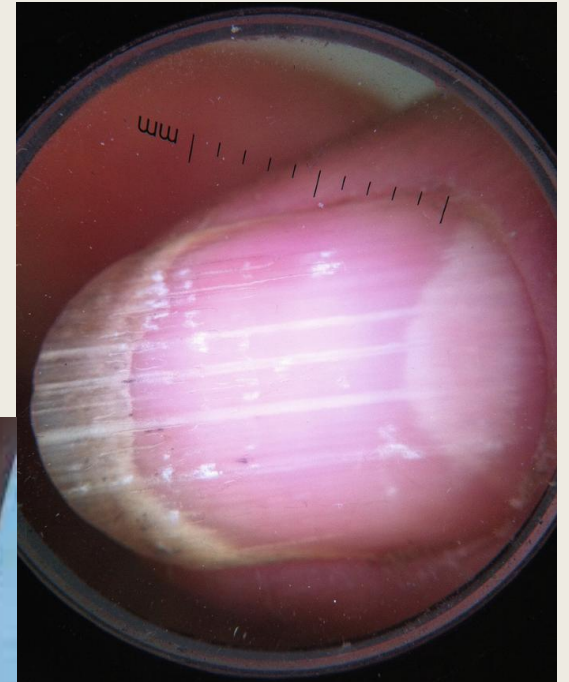
Clinical Presentation

- Mild *flaccid vesicles* on erythematous skin (often not noticed) → *then progress* to large, macerated, exudative plaques of superficial erosions with crusting



Clinical Presentation

- *Longitudinal, white bands of nails*
(in 70% of pts)



Location

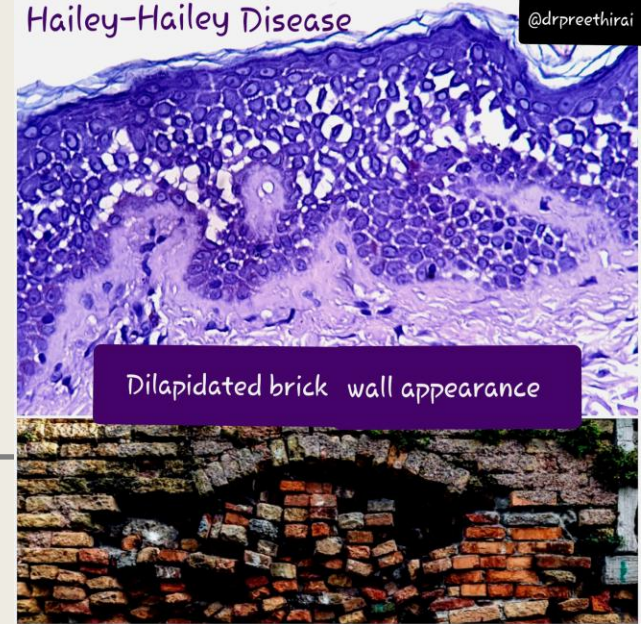
- *Flexural areas* (groin, axillae, neck, submammary, perineum)
- *Symmetric*
- Rarely mucosal



Complications

- Remitting and relapsing course
- *Secondary* bacterial (staph, strep) and fungal (candida) → clues of infection include malodor and vegetations → culture sample → tx with abx or antifungal
- Anecdotal skin cancer risk

Diagnosis



- Biopsy and FH
- **Punch biopsy Biopsy** (4mm)
 - Ideal:
 - *Punch* biopsy to get all layers
 - Early lesion
 - 2 samples or Edge of blister/erosion (to get BOTH normal and abnormal skin)
 - Abnormal portion → H&E staining → acantholysis of intraepidermal and suprabasal keratinocytes ("dilapidated brick wall")
 - Normal portion → direct immunofluorescence = **negative**
- Lab
 - Referral to Genetics for **genetic testing and counseling**
 - ELISA = **negative**
 - Indirect immunofluorescence = **negative**
 - Fungal and bacterial cultures if suspect secondary infection

Treatment

- No specific therapy
- Goal: control exacerbating factors, secondary infections, and cutaneous inflammation
- *Due to rarity, topical/systemic therapy based on small observational studies, case reports, and clinical experience
- Mild → 1st line = topical antimicrobials (ex: clindamycin) and topical steroids
- Severe → Intralesional botulinum toxin type A (to reduce sweating and maceration), systemic abx, DMARDs (cyclosporin, methotrexate)

Treatment

- General measures (heat, moisture, friction provoke cutaneous flares)
 - Cool, dry environment
 - Lightweight, loose clothing
 - Absorbent pads in skin folds
 - Reduction of body weight, if appropriate

Treatment

- General measures (to prevent secondary infections)
 - Open, crusted, macerated, malodorous lesions can be treated with diluted bleach → apply as baths, soaks, compresses 2-3x per week
 - Other alternatives (antibacterial washes ex: chlorhexidine gluconate 4% wash)

***Bullous
pemphigoid***

Causes / Population

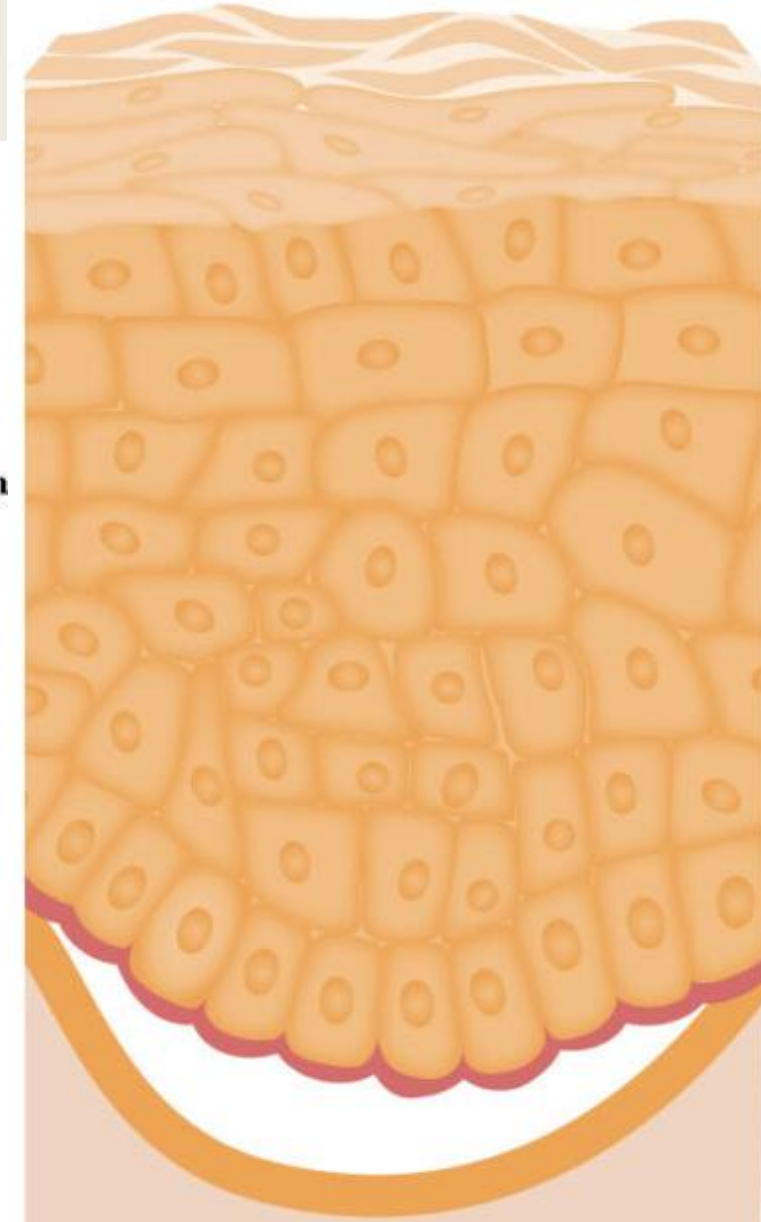
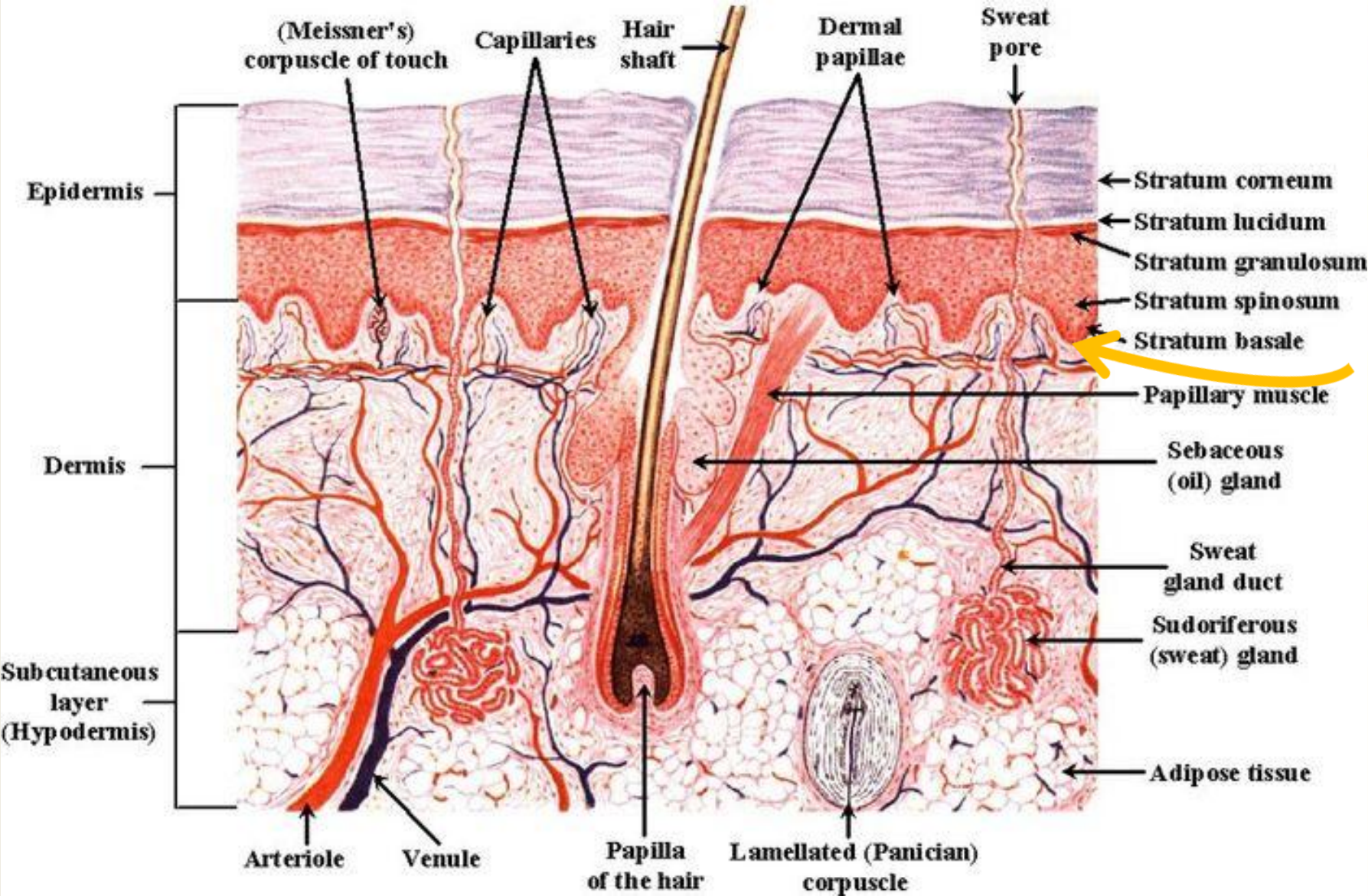
- Causes
 - Can be induced medications (ex: DPP-4 inhibitors, furosemide, captopril, abx) → discontinuation results in improvement
 - May be linked to pts with neurologic conditions
- Population
 - >60yo (*older adults*)
 - male=female

IgG Auto-Antibodies to:

- *Hemidesmosomal Proteins*
 - *BP180 (BP antigen 2)*
 - *BP230 (BP antigen 1)*

Layer that separates

- *SUB epidermis*



Clinical Presentation

- ***Prodrome*** of pruritis and skin lesions (papular, eczematous plaques, urticaria-like) for weeks to months
- ***Tense bullae***



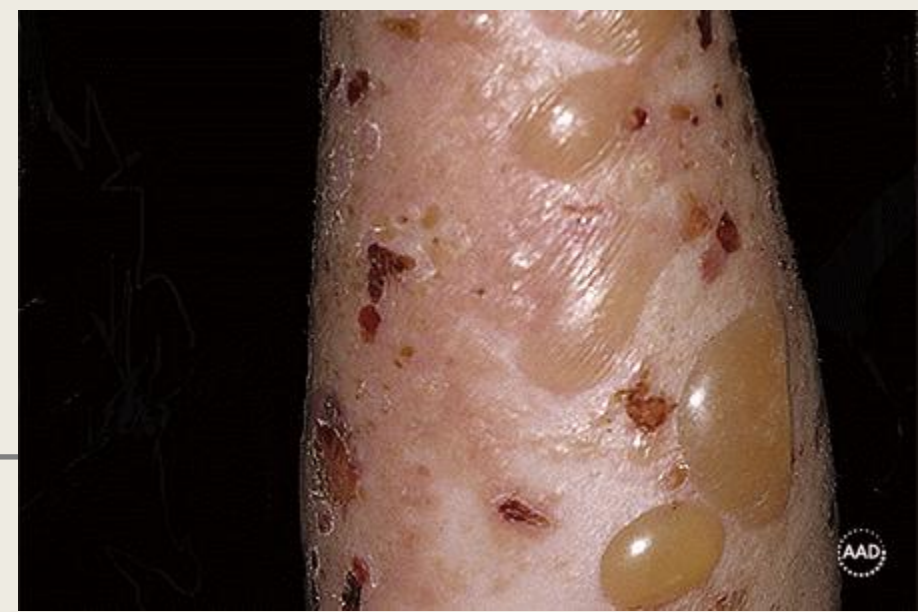
Clinical Presentation

- 1-3cm tense bulla



Clinical Presentation

- Eventually, the bullae rupture, leaving moist erosions and crusts



Location



- *Cutaneous > mucosal*
- *Trunk, extremity flexures, axillary and inguinal folds... if mucosa, would involve mouth*
- *Nail changes*



Signs

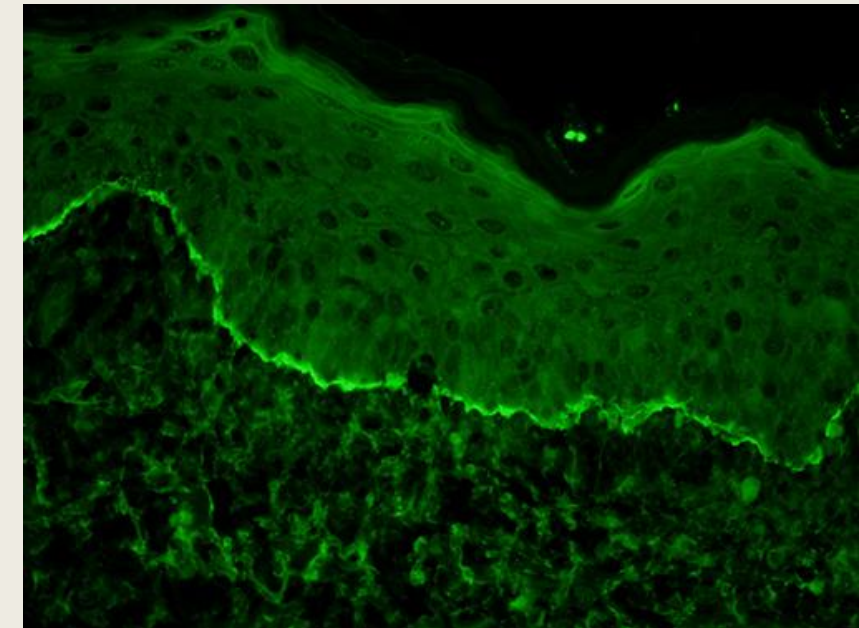
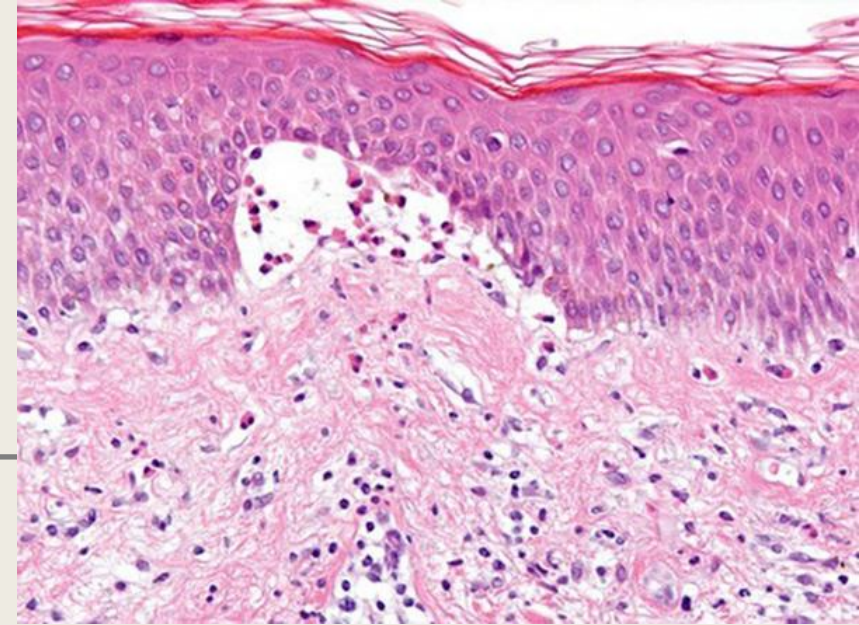
- (-) Nikolsky's sign

Complications

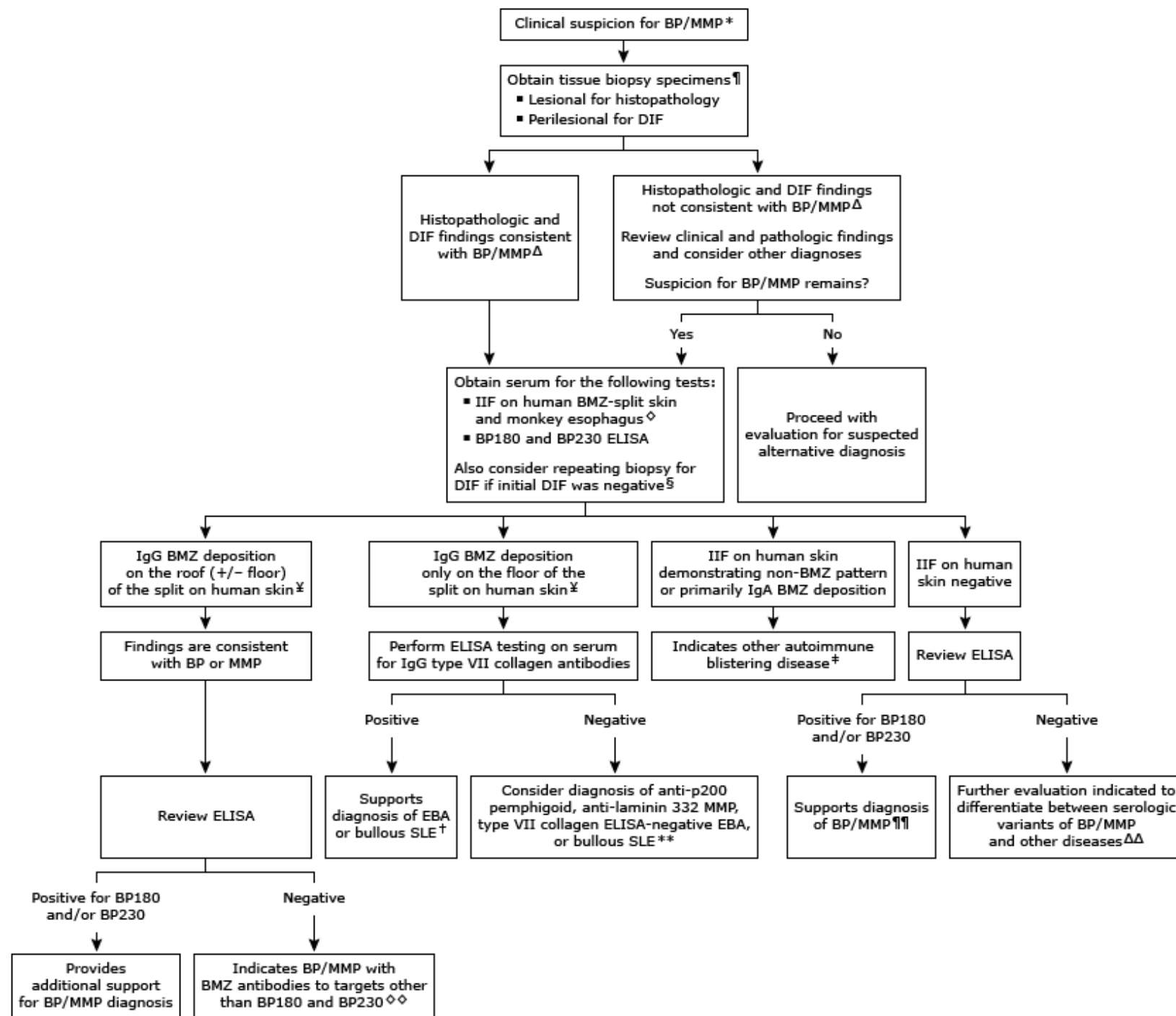
- Potentially fatal
- Heart disease, infection, neurological disease

Diagnosis

- ***Older patient, (-)Nikolsky sign, unexplained pruritis at first*
- **Punch biopsy** (two 4mm biopsies)
 - Ideal:
 - *Punch* biopsy to get all layers
 - Abnormal portion → H&E staining → *epithelial-subepithelial cleavage*
 - Normal portion → direct immunofluorescence showing *linear deposition of IgG along basement membrane*
- Lab
 - ELISA
 - *+ IgG abs to BP180 and BP230*
 - Indirect immunofluorescence
 - Human basement membrane zone-skin and monkey esophagus substrates



Diagnosis



Treatment

- Goal
 - Decrease blister formation, promote healing, improve quality of life
- Pretreatment considerations
 - *Med induced? → stop the med*
 - *Older patients → increased medication serious adverse effects and immunosuppression*
- Approach
 - Treat at time of disease onset, even if mild
 - *Conservative, careful approach to treatment regimen: Morbidity and mortality result from adverse effects of treatment.*

Treatment

- Initial
 - Choose one of the following:
 - High-potency *topical steroids* BID (ex: clobetasol)
 - *oral steroids* (ex: prednisone) – **rapid symptoms control**
 - doxycycline (slower remission than steroids, but more favorable side effect profile)
- Extensive (widespread or continued development of many new bullae per day)
 - Add DMARD
 - Conservative, careful approach to treatment regimen: Morbidity and mortality result from adverse effects of treatment.
- Refractory
 - Biologics/monoclonal Ab (ex: rituximab) and IVIg

Treatment

- End of consolidation phase (no new lesions or pruritic symptoms for at least 2 weeks)
- Slow taper (ex: 5mg reduction of prednisone dose q2-4 weeks)

Examples of doses considered minimal therapy include:

- [Clobetasol](#) propionate – ≤ 20 g of clobetasol propionate per week
- [Prednisone](#) – ≤ 0.1 mg/kg per day
- [Azathioprine](#) – ≤ 0.7 mg/kg per day (patients with normal thiopurine S-methyltransferase [TPMT] level)
- [Mycophenolate](#) mofetil – ≤ 500 mg per day
- Mycophenolic acid – ≤ 360 mg per day
- [Methotrexate](#) – ≤ 5 mg per week
- [Dapsone](#) – ≤ 50 mg per day
- [Doxycycline](#) – 100 mg twice daily

Treatment

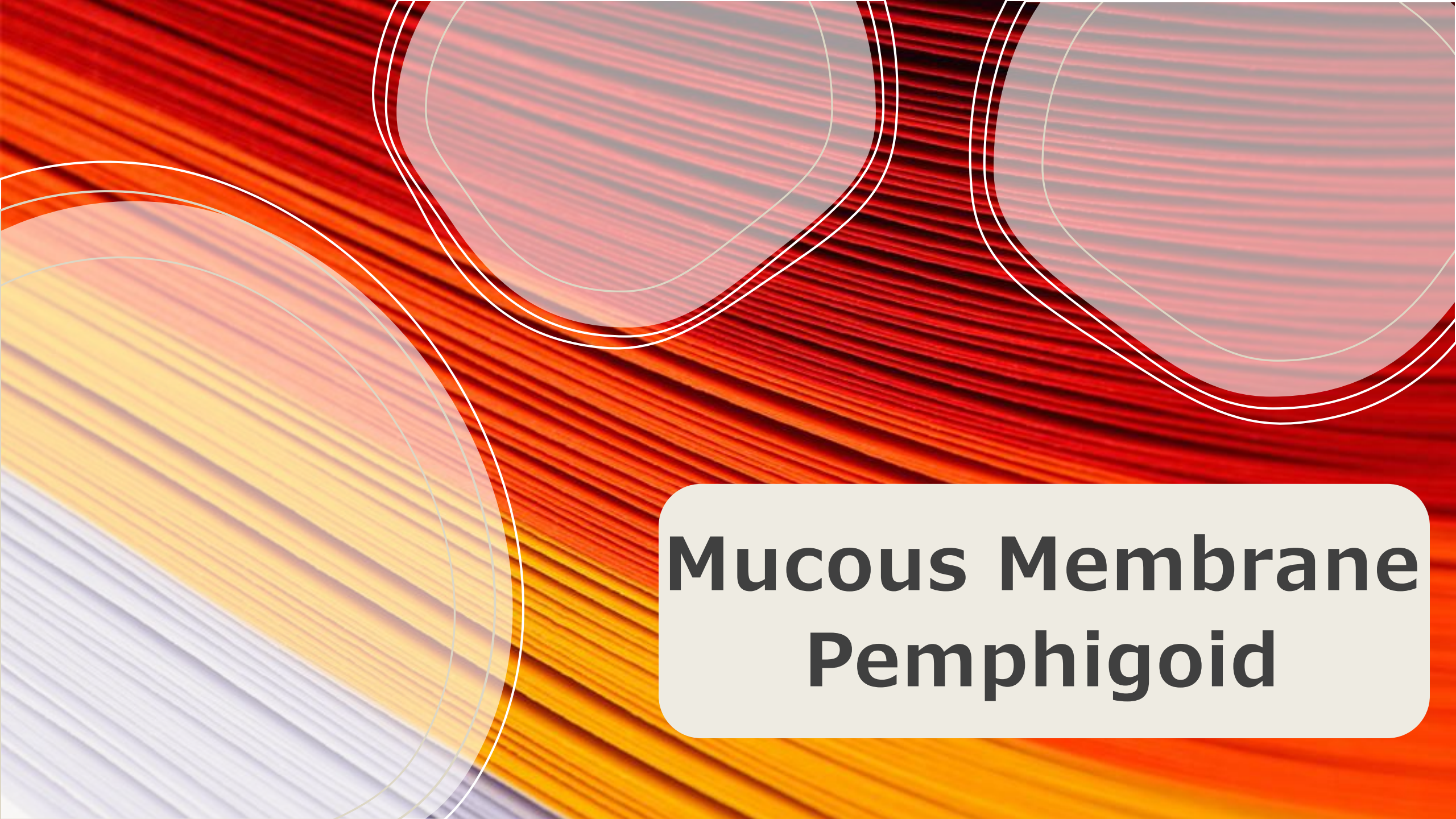
- General skin care
 - Daily rupturing of tense blisters reduce lateral extension of blister edges
 - Clean blisters with alcohol → puncture in most gravity dependent area with sterile needle/blade → leave blister roof in place (acts as natural wound dressing) → cover open erosions with nonstick dressing
 - Education on infection precautions
- If oral involvement
 - Topical steroids

Treatment

- Referrals
 - Dermatology
 - If vision changes, itching/burning sensation of eyes → Ophthalmology
 - Dysphagia/esophageal symptoms → Gastroenterology
 - Hoarseness, difficulty speaking, dysphagia, nasal obstruction symptoms → Otolaryngology
 - [International Pemphigus and Pemphigoid Foundation](#)

Prognosis

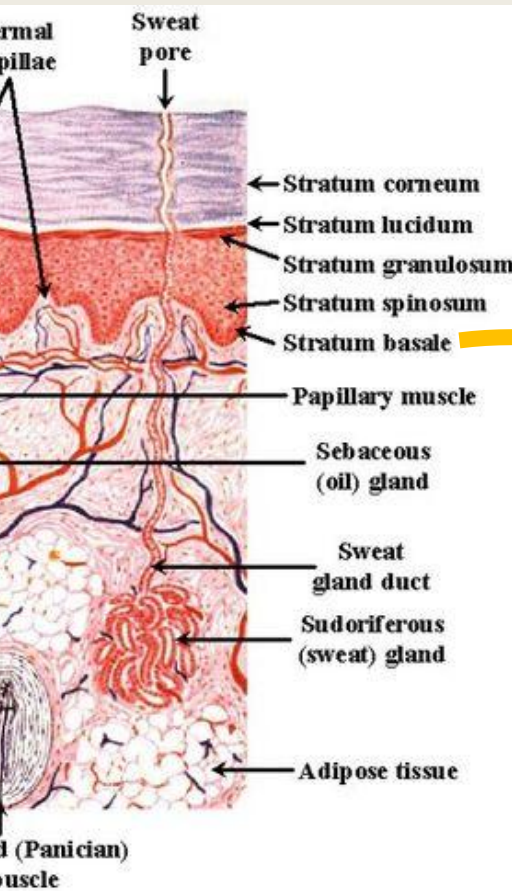
- Chronic, relapsing
- Long-term remission may occur after months to years
- If drug-induced → most often has limited course after cessation of inciting drug
- 1 year overall mortality 11-48%



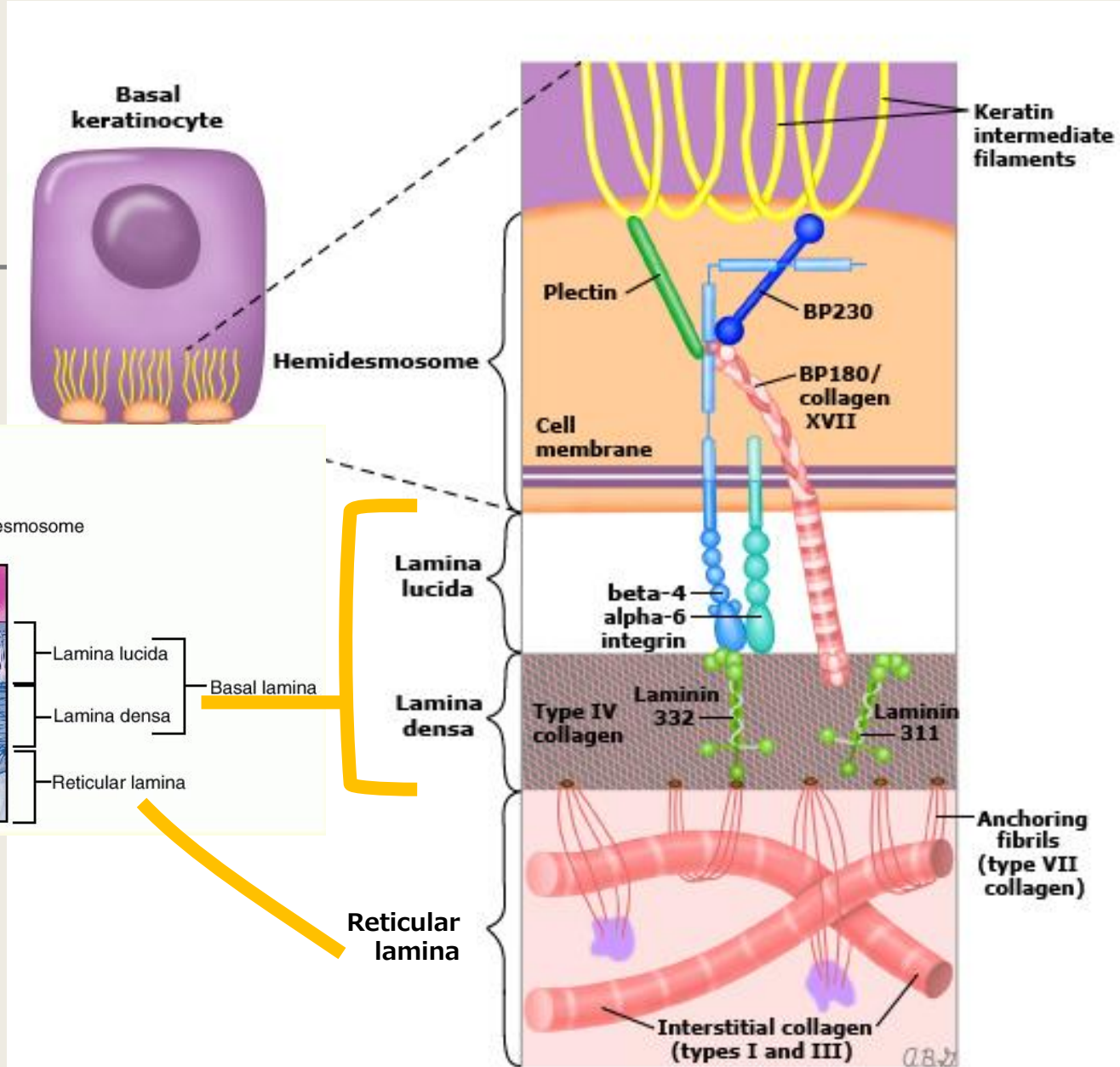
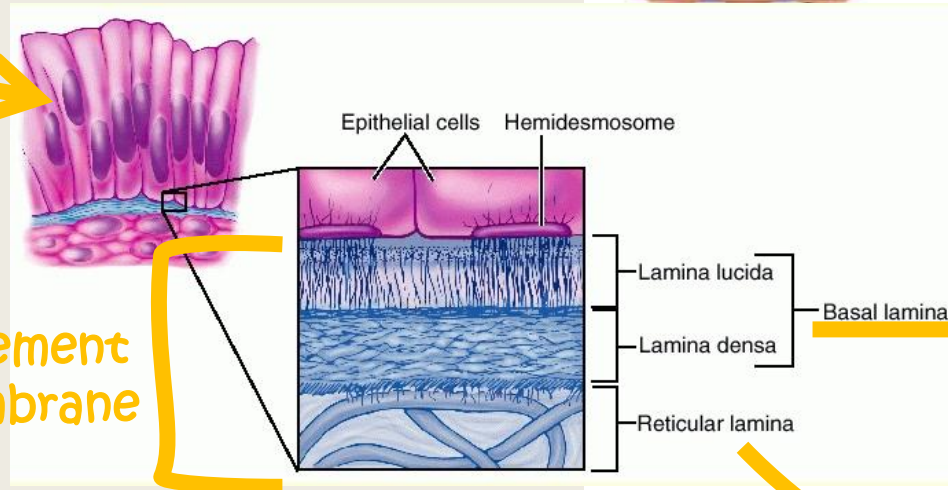
Mucous Membrane Pemphigoid

Causes / Population

- Causes
 - Like Bullous pemphigoid, Can be medication induced
- Population
 - >60yo (older adults)
 - 2 per million cases per year

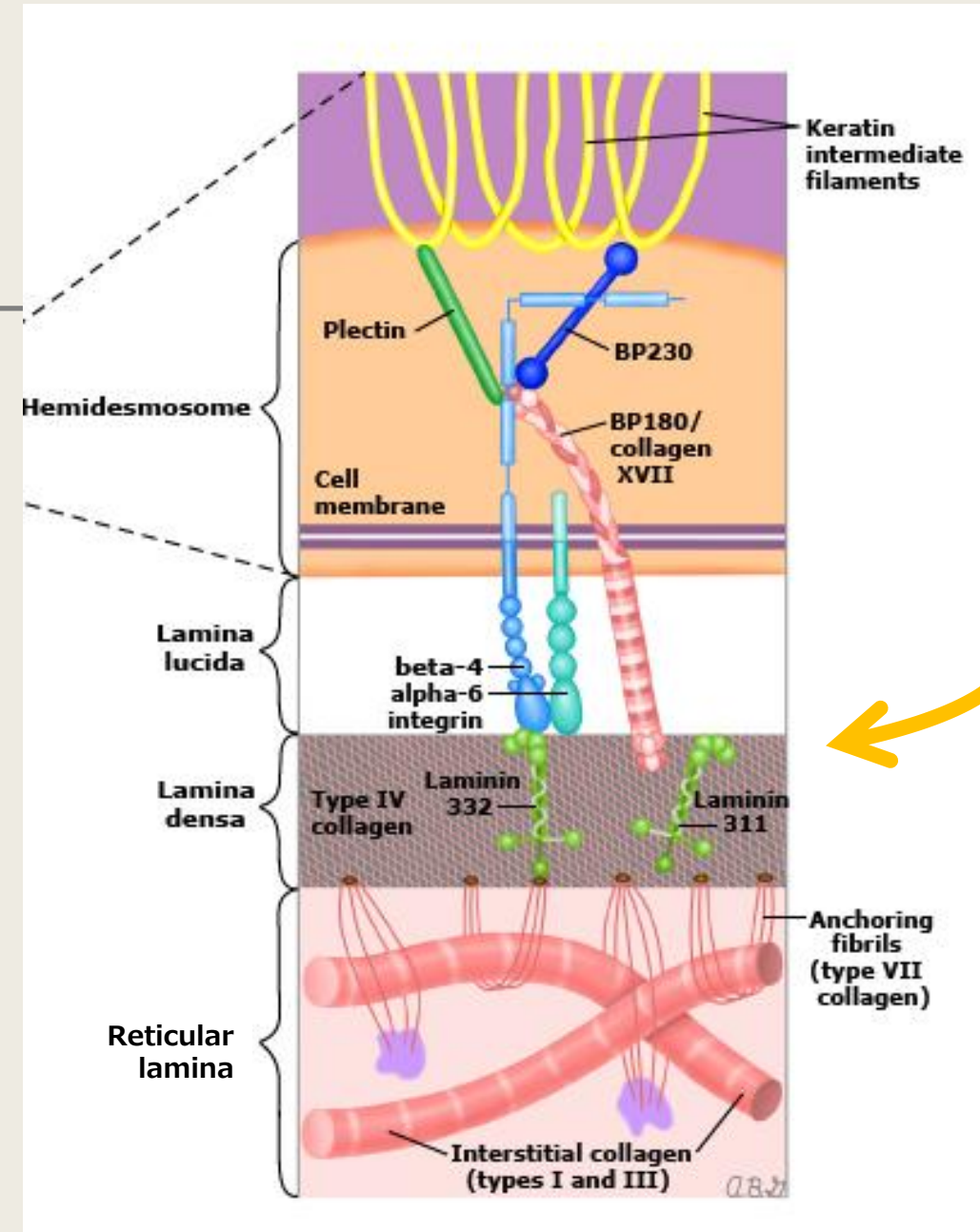


Basement membrane



IgG Auto-Antibodies to:

- In contrast to bullous pemphigoid, Ab finding is located in lower lamina lucida and lamina densa of basement membrane → increase risk of scarring
 - BP180 (BP antigen 2)
 - BP230 (BP antigen 1)
 - Laminin 332
 - Alpha-6 beta-4 integrin
 - Type VII collagen



Clinical Presentation and Location



- Erosions (don't usually see the bullae)
- **Mucosal sites** of head and upper trunk (oral cavity and ocular conjunctival most common)
- Mucosal scarring can lead to blindness, airway obstruction, and dysphonia
- Nail changes



Diagnosis

- *Older patient, desquamative mucosal involvement, unexplained pruritis*
- Similar to Bullous pemphigoid
- Punch biopsy Biopsy, ELISA, indirect immunofluorescence
 - See antibodies along **DERMAL SIDE of the split**
 - + ab to Laminin 332, Alpha-6 beta-4 integrin, Type VII collagen, BP180, BP230,

Treatment

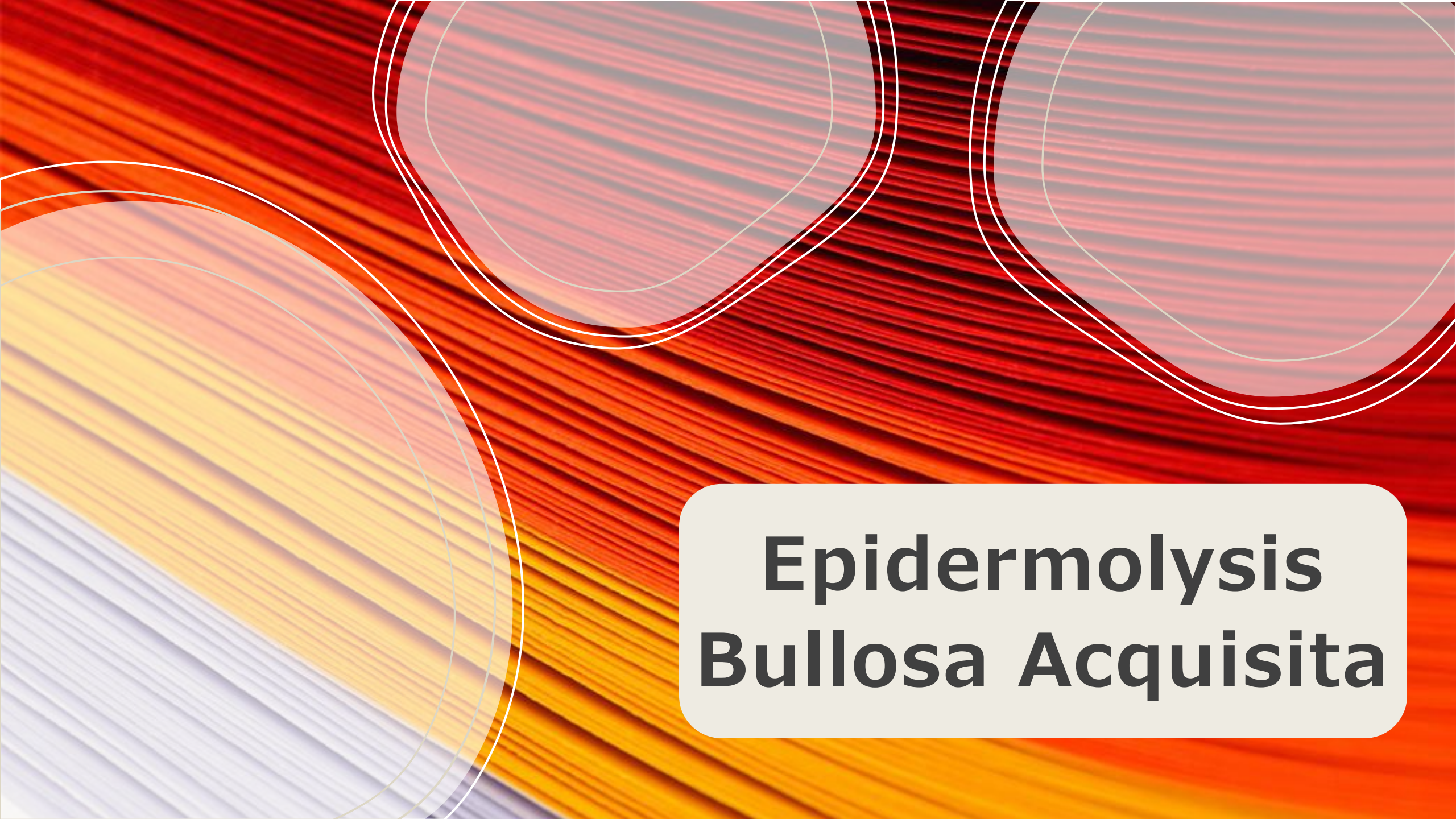
- Goal
 - Halt progression of disease, improve symptoms, prevent adverse sequelae of chronic tissue inflammation and scarring

Treatment

- Mild, localized to oral cavity
 - First dry affected area with tissue → 1st line= topical super high potency steroids (clobetasol ointment/gel BID-QID)
→ avoid eating/drinking for 30 min
 - Other options: topical tacrolimus
- Moderate
 - Add Systemic steroids and dapsone
- Severe
 - Biologics/monoclonal Ab (ex: rituximab) and IVIg

Treatment

- Nasal Involvement
 - Saline nasal irrigation BID, humidifier, nasal emollients, steroid nasal spray
- Esophageal involvement
 - Soft diet, GERD tx, esophageal dilation of strictures
- Anogenital involvement
 - Topical steroids or calcineurin inhibitors, stool softeners, dilation
- Ocular involvement
 - Lubrication, lid hygiene, trichiasis management



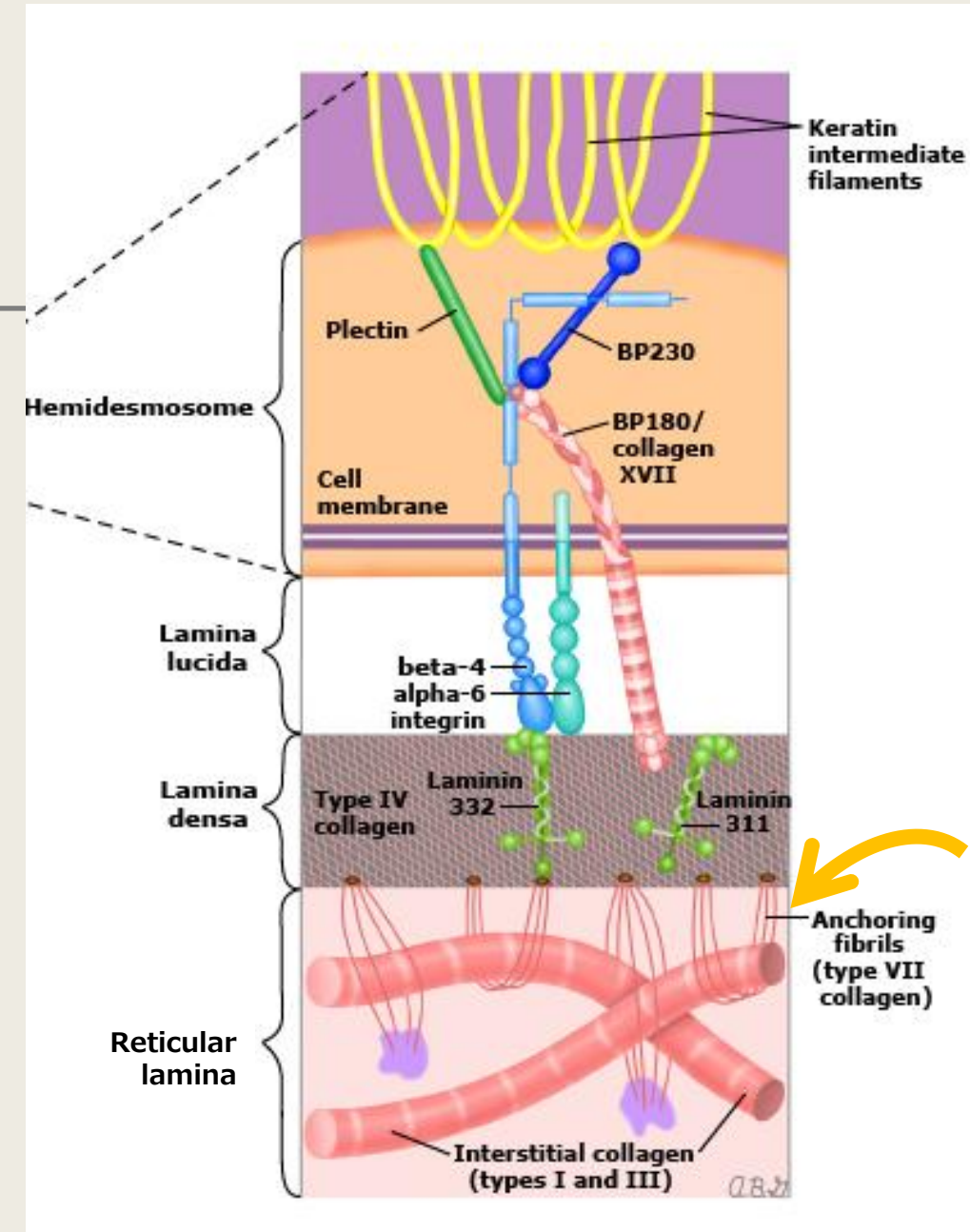
Epidermolysis Bullosa Acquisita

Population

- Triggers
 - Secondary event following skin injury
 - Possible genetic
- Population
 - adults

IgG Auto-Antibodies to:

- **Type VII collagen** (connects basement membrane to dermal structures)

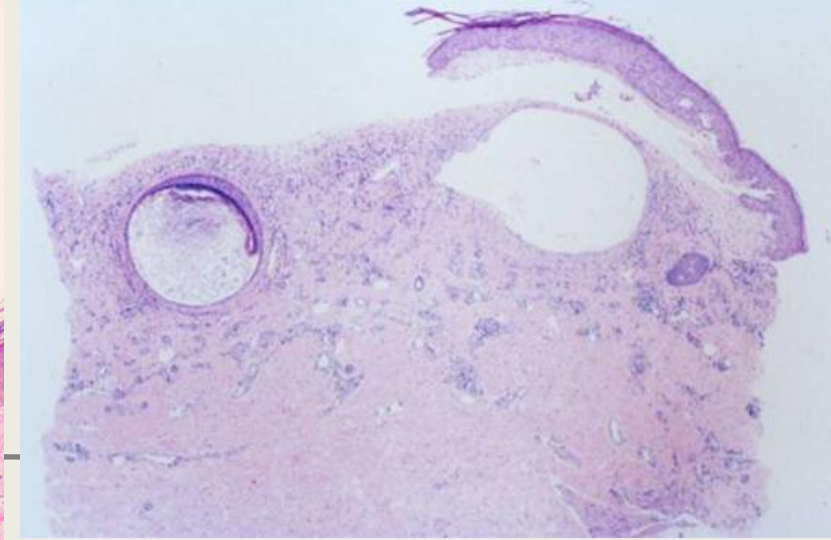
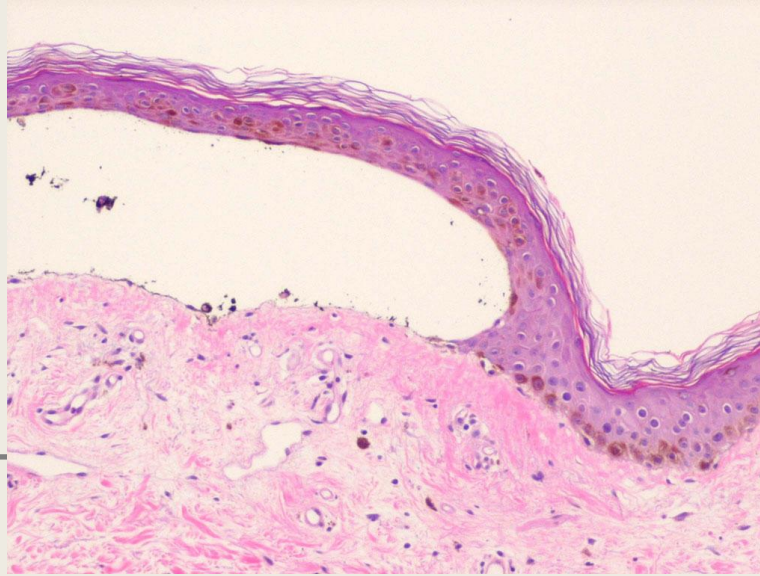


Clinical Presentation and Location

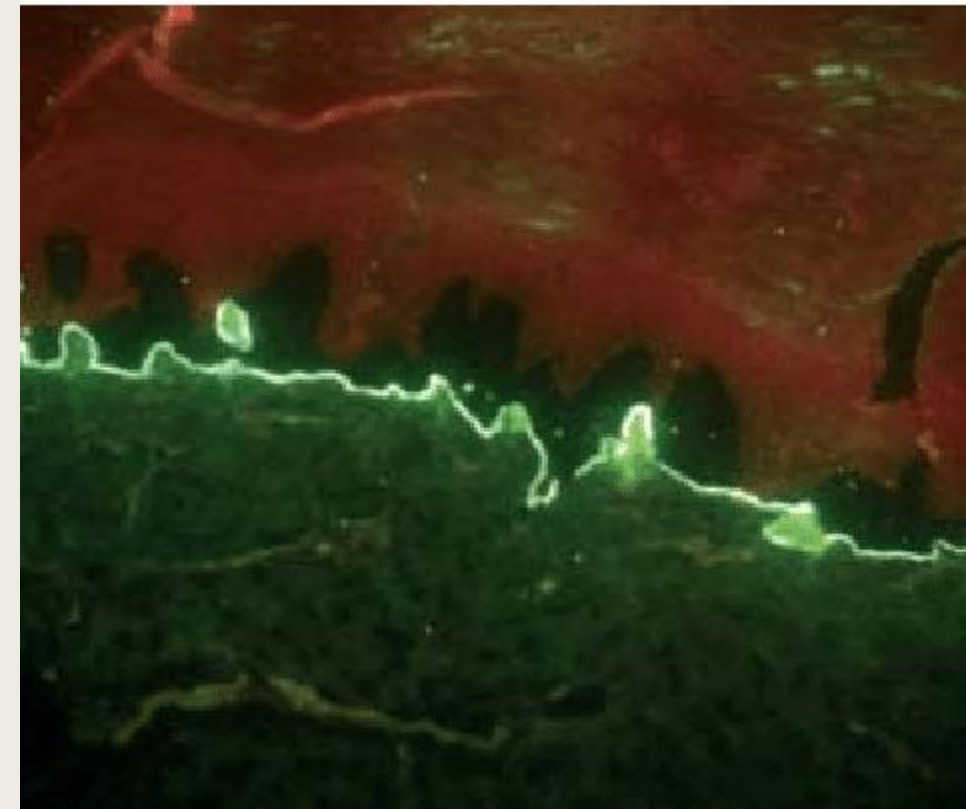
- Tense bullae
- mild mucosal involvement
- healing with dense scars
- *primarily at trauma-prone areas*
(hands, feet, knees, elbows, buttocks)



Diagnosis



- Two 4mm **Punch biopsies**, ELISA, indirect immunofluorescence of salt-split skin
 - **Subepithelial blister**
 - Bind to **Type VII collagen**
 - **Linear IgG that binds to dermal floor** (lower part)



Treatment

- Notoriously difficult to treat, refractory
- Goal
 - In addition to pharmacologic therapy, minimize disease severity

Treatment

- Initial
 - *Colchicine or dapsone*
 - Classical EBA may be less likely to respond to systemic steroids, so don't start with steroids
- Severe
 - Add systemic steroids to colchicine/dapsone
- Refractory
 - Rituximab, IVIG

Treatment

- General measures
 - Avoid trauma to the skin → gentle cleansing of skin during bathing and avoid scrubbing and harsh soaps
 - Infection recognition education
- Wound care
 - Hydrogel or semioclusive hydrocolloid dressings in new, clean wounds
 - Silver impregnated dressing q48 hours to minimize infection risk
- Involve ophthalmology, gastroenterology, dentist, for respective involved areas



Dermatitis Herpetiformis

Population

- Triggers
 - Associated with gluten-sensitive enteropathy (*celiac disease*)
- Population
 - 20-50yo
 - Male>female
 - Northern European descent

IgA Auto-Antibodies to:

- *tissue transglutaminase*

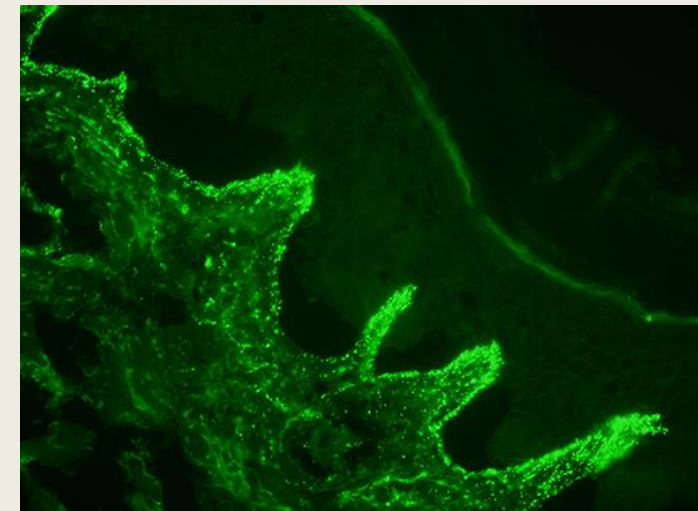
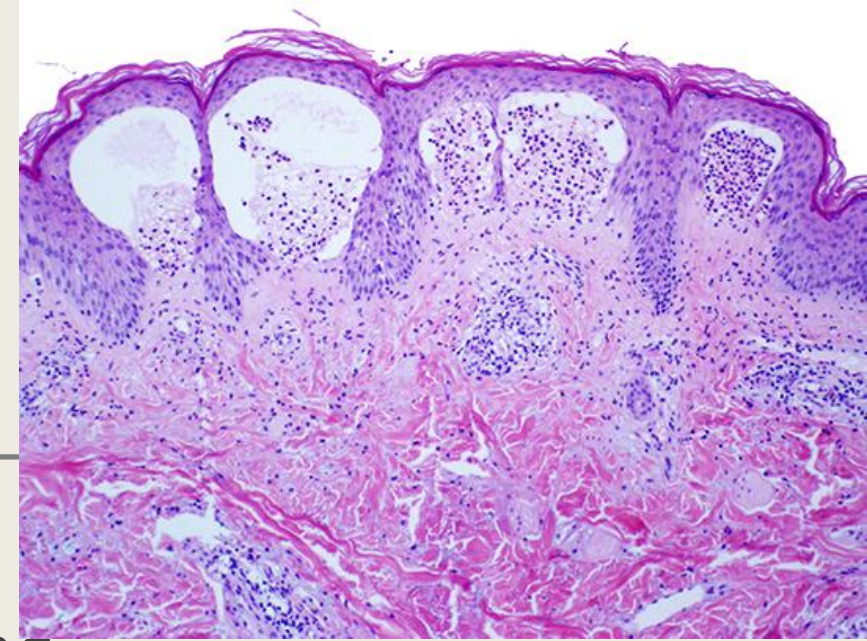
Clinical Presentation and Location

- Extremely *pruritic clusters of urticarial papules or vesicles*
- Symmetric on extensor surfaces
- Extremities, buttock, scalp, back. Mucous membranes spare
- Only excoriations and hemorrhagic crusts may be seen due to scratching from intense pruritis
- Associated with Hashimoto thyroiditis and enteropathy associated T cell lymphoma



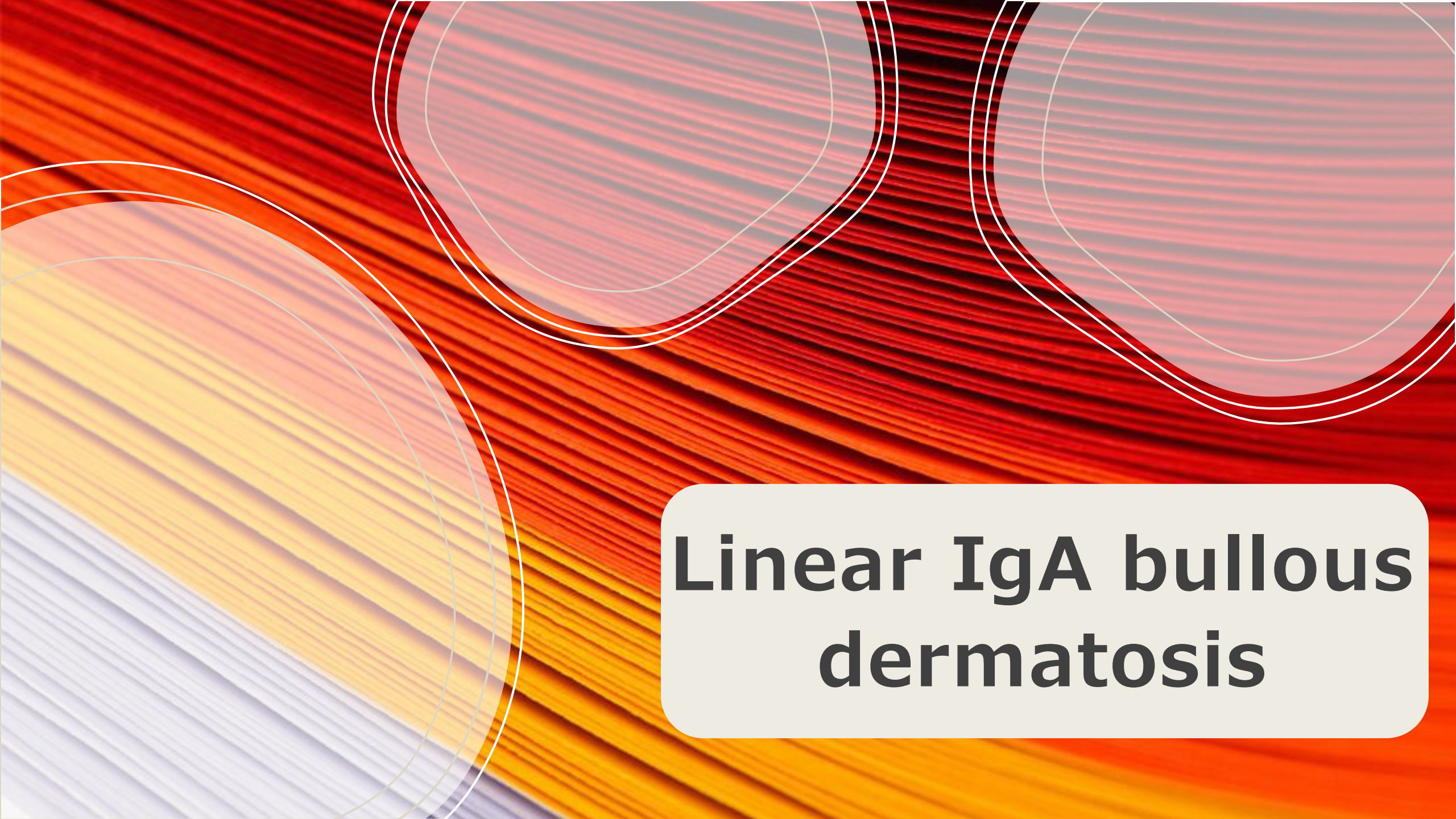
Diagnosis

- Two 4mm Punch biopsies and serology
 - Specimen Histology: subepidermal blistering, neutrophilic microabscesses, perivascular inflammatory infiltrate
 - Specimen DIF: gold standard → IgA deposits within dermal papillae
 - Serology: ELISA IgA tissue transglutaminase abs Serology: Indirect immunofluorescence for IgA endomysial Ab
 - (serum Ab levels fall with gluten restriction)



Treatment

- Referral to **Gastroenterologist**, dermatologist, nutritionist
- *1st line= Gluten restriction + Dapsone*
 - Taper dapsone to minimum dose that suppresses pruritis and new lesions after pt has been on strict gluten-free diet for at least 2-3 months



Linear IgA bullous dermatosis

Population

- Triggers
 - Idiopathic
 - Maybe meds in adults, esp: *vancomycin*
 - Associated with *ulcerative colitis*
- Population
 - Adults >60 yo
 - Children 6mo-10yo

IgA Auto-Antibodies to:

- 97 kDa antigen
- 120 kDa antigen
- laminin-332
- BP180 (NC16a domain)
- (on epidermal side of dermoepidermal junction)

Clinical Presentation and Location



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- Cutaneous & mucosal
- *Tense bullae in annular distribution or linear*

Or

- Targetoid lesions whose outer rim is an annular bulla
→ *crown of jewels*



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Clinical Presentation and Location



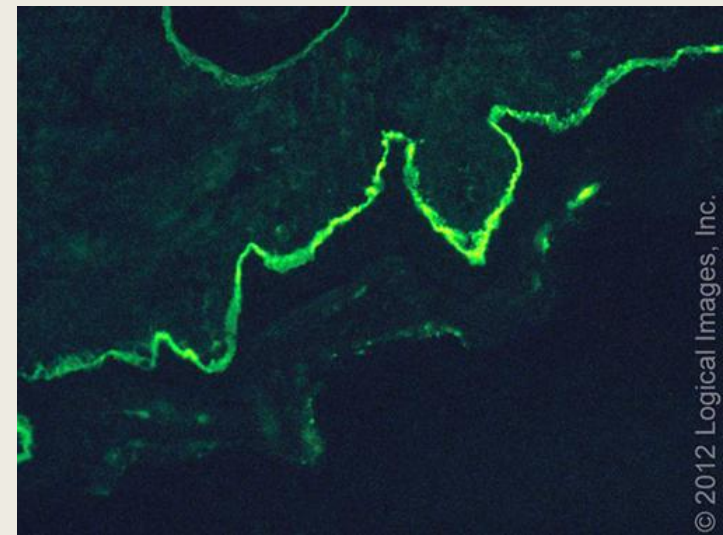
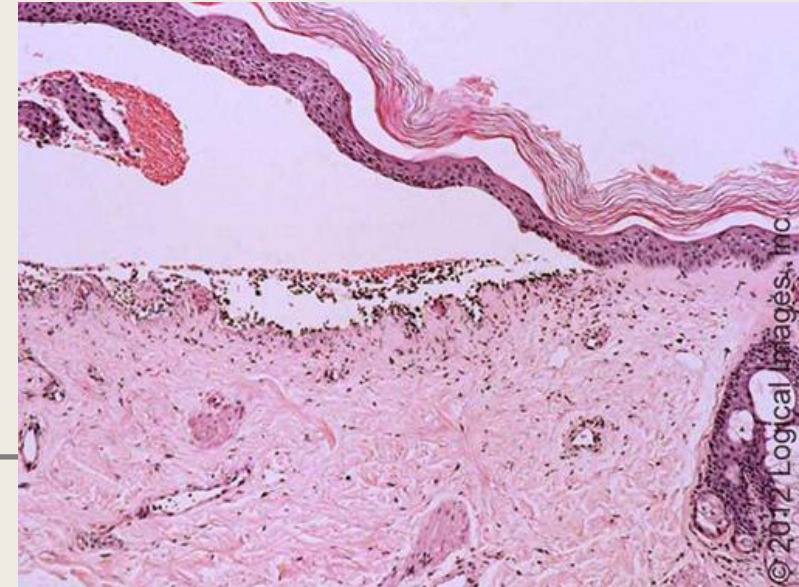
- Mucosal lesions primarily present as **erosions or ulcers** (uncommon to see intact bullae)
- If involve mucosa, oral and ocular are most common



Diagnosis

- Two 4mm **Punch biopsies** and serology
 - Specimen Histology: subepidermal blistering, neutrophilic inflammatory cell infiltrate
 - Specimen DIF: **IgA deposits along basement membrane (both dermal and epidermal side)**
 - Serology: **Indirect immunofluorescence for 97 kDa, 120 kDa, laminin-332, and BP180, Type VII collagen (COL7), laminin-332, or laminin gamma**

1

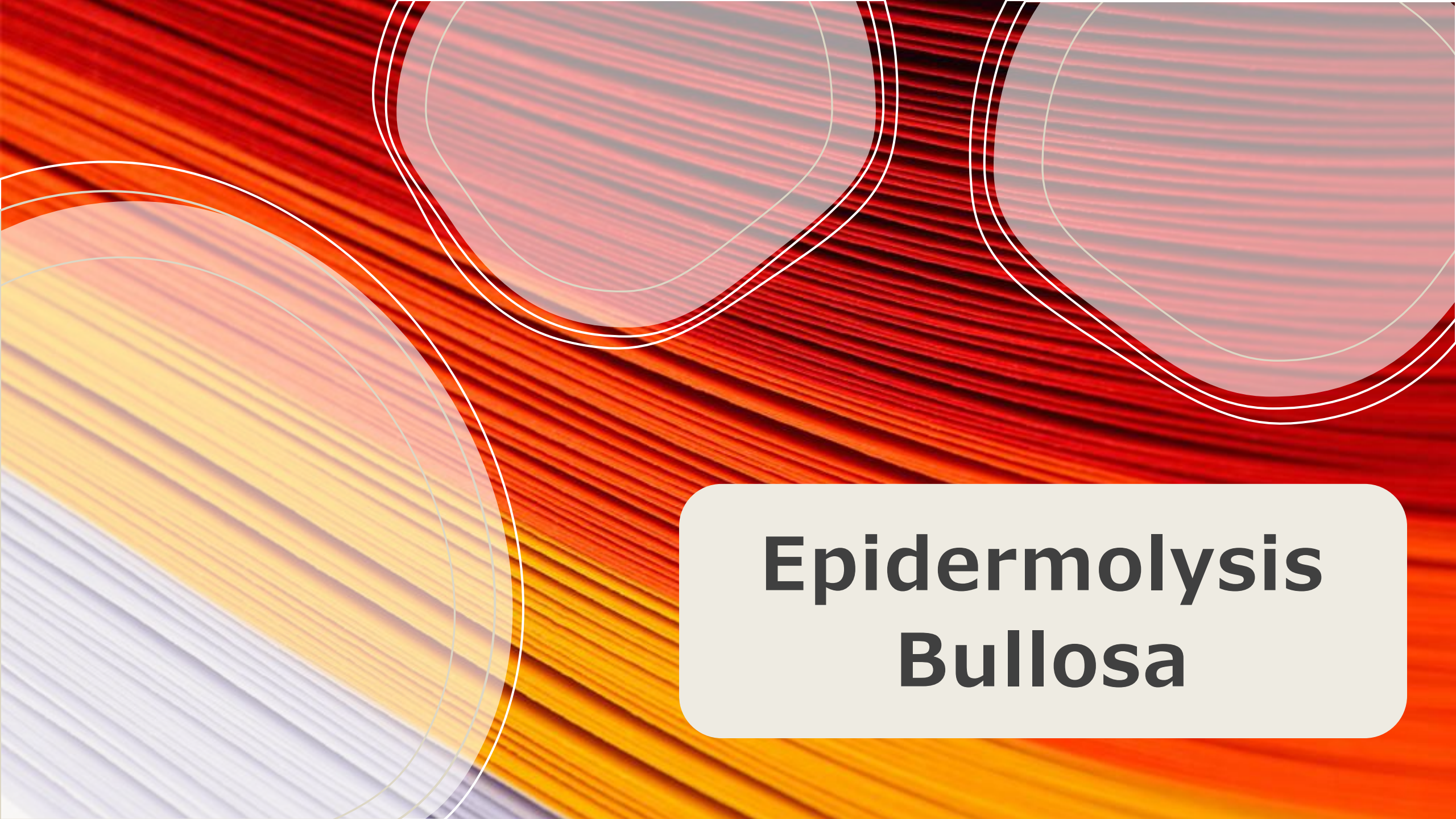


Treatment

- If drug-induced, stop the drug
- *1st line= Dapsone*
 - Start low, titrate up
 - Also, topical steroids
- severe and refractory
 - Systemic steroids, DMARDs, biologics, IVIG

Prognosis

- Children = typically self-limited, lasting avg 2-4 years
- Adult = chronic (months to years, even decades)



Epidermolysis Bullosa

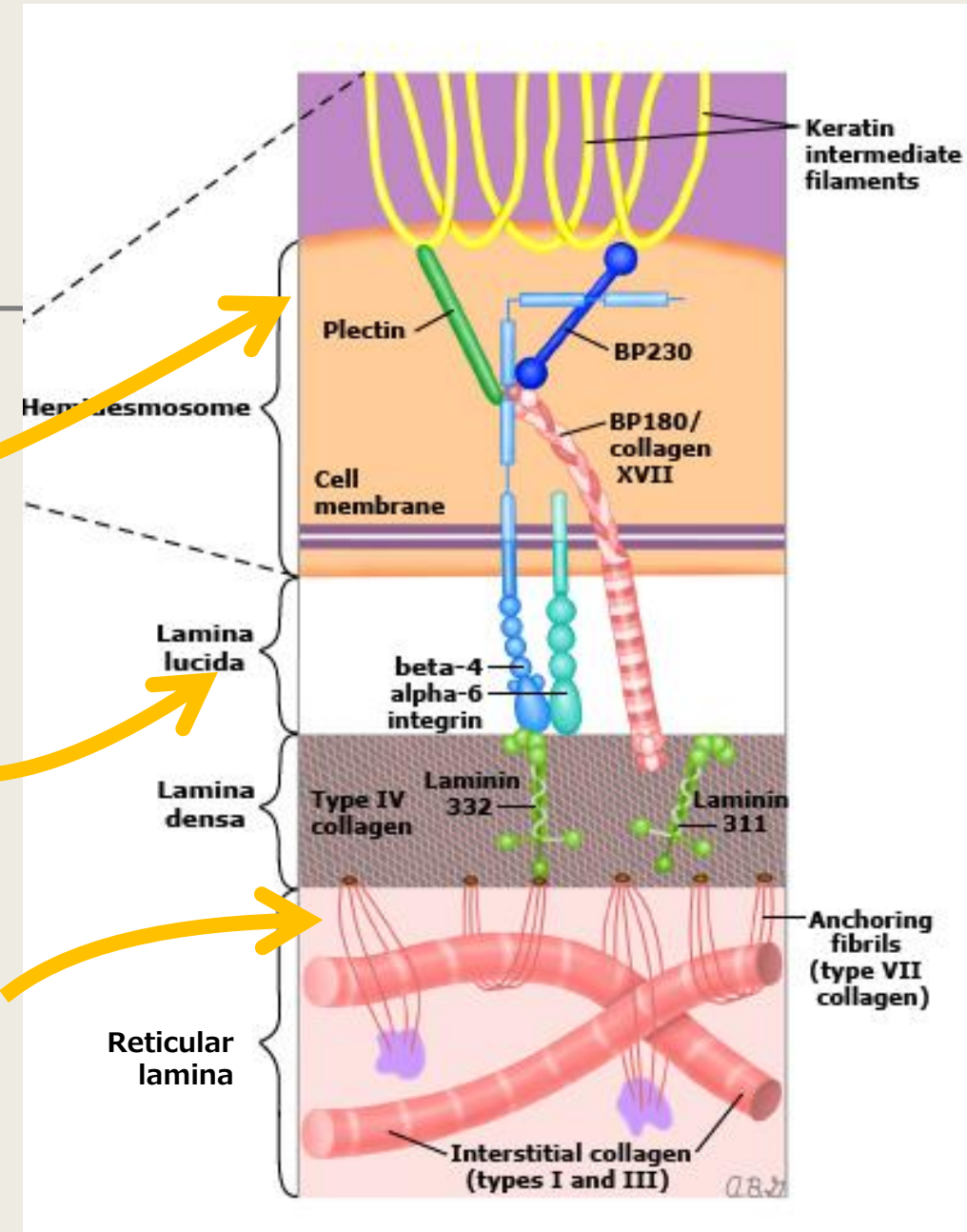
Population

- Cause
 - *Genetic defect*
- Population
 - encompasses a group of extremely rare inherited bullous diseases

Layer that separates

3 major forms of Epidermolysis bullosa:

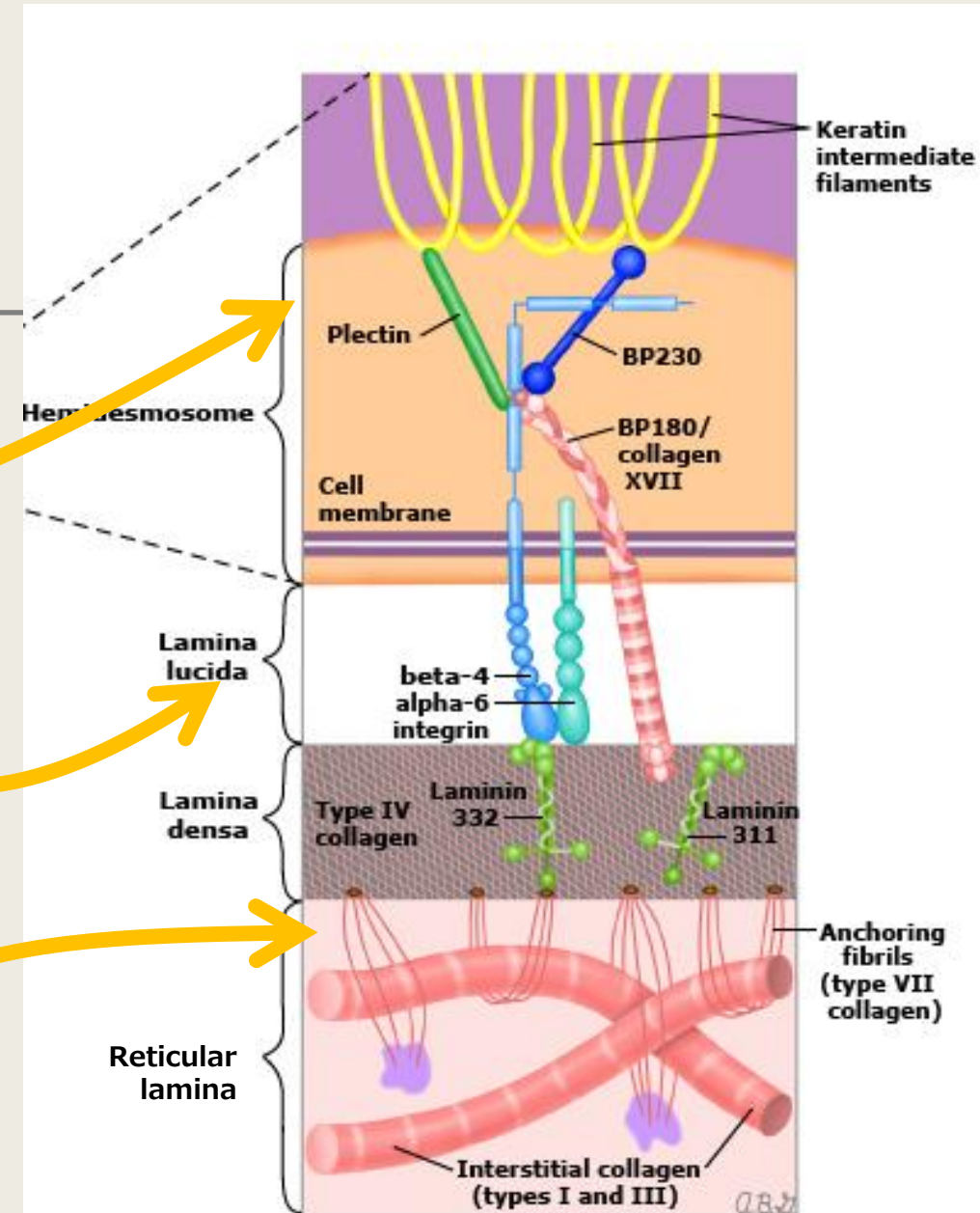
- Epidermolysis bullosa simplex (EBS)
 - Within basal cell layer
- Junctional epidermolysis bullosa (JEB)
 - Within lamina lucida
- Dystrophic epidermolysis bullosa (DEB)
 - Within sublamina densa



Inherited Protein defect

3 major forms of Epidermolysis bullosa:

- Epidermolysis bullosa simplex (EBS)
 - Desmoplakin, plakophilin-1, **keratin 5**, **keratin 14**, plectin
- Junctional epidermolysis bullosa (JEB)
 - **Laminin 332**, collagen XVII, alpha 6/beta4 integrin
- Dystrophic epidermolysis bullosa (DEB)
 - **Collagen VII**



Clinical Presentation and Location - Epidermolysis bullosa simplex (EBS)

- Blister, erosions, crust
- *Formation results from shearing force*
- Ranges from **minor** blistering to **lethal** outcome



Clinical Presentation and Location - Junctional epidermolysis bullosa (JEB)

- Blistering of skin and mucosae that heal with scarring
- *Mild at birth → rapidly progresses, can be associated with early lethality*
- Cutaneous, mucosal, dental abnormalities, upper airway involvement, urologic involvement



Clinical Presentation and Location - Dystrophic epidermolysis bullosa (DEB)

- Skin fragility, blistering, scarring, nail changes, milia formation in areas of healed blistering
- Mild= localized to toenails
- *Severe= blistering and scarring that leads to fusion of fingers and toes*



Complication

- Malnutrition and anemia (transcutaneous loss of nutrients, malabsorption, feeding difficulties)
- Infection
- Skin cancer

Diagnosis

- Skin biopsy
- Genetic testing

Treatment – supportive care

- *Avoid friction (ex: don't open twist-cap bottles)*
- Bath with salt water that approximates isotonic saline
- Nonadherent silicone dressing, foam dressings
- Novel therapy → topical gene therapy
- *Drain blisters in sterile fashion* and leave roof intact
- Bleach baths/chlorhexidine
- Pain management
- Nutritional support (may need G-tube)
- Monitor micronutrients, multivitamin supplementation

Treatment – supportive care

- Consult to Epidermolysis Bullosa center
- Genetic consult



Infectious Vesicobullous Diseases

Staph Bullous Impetigo

- Staph aureus → exfoliative toxin
→ cleaves desmoglein 1
- Localize
 - can be treated with topical abx
(mupirocin TID x5 days)
- Scalded skin syndrome → from hematogenous spread of the exfoliative toxin
 - Immediate admission to hospital
 - IV abx



Disseminated Herpes Simplex

- Most commonly on eczematous skin of atopic dermatitis pts, resulting in eczema herpeticum
- can also occur in severely immunocompromised pts (2/2 HIV or iatrogenically)
- Tx: antiviral (ex: valacyclovir)

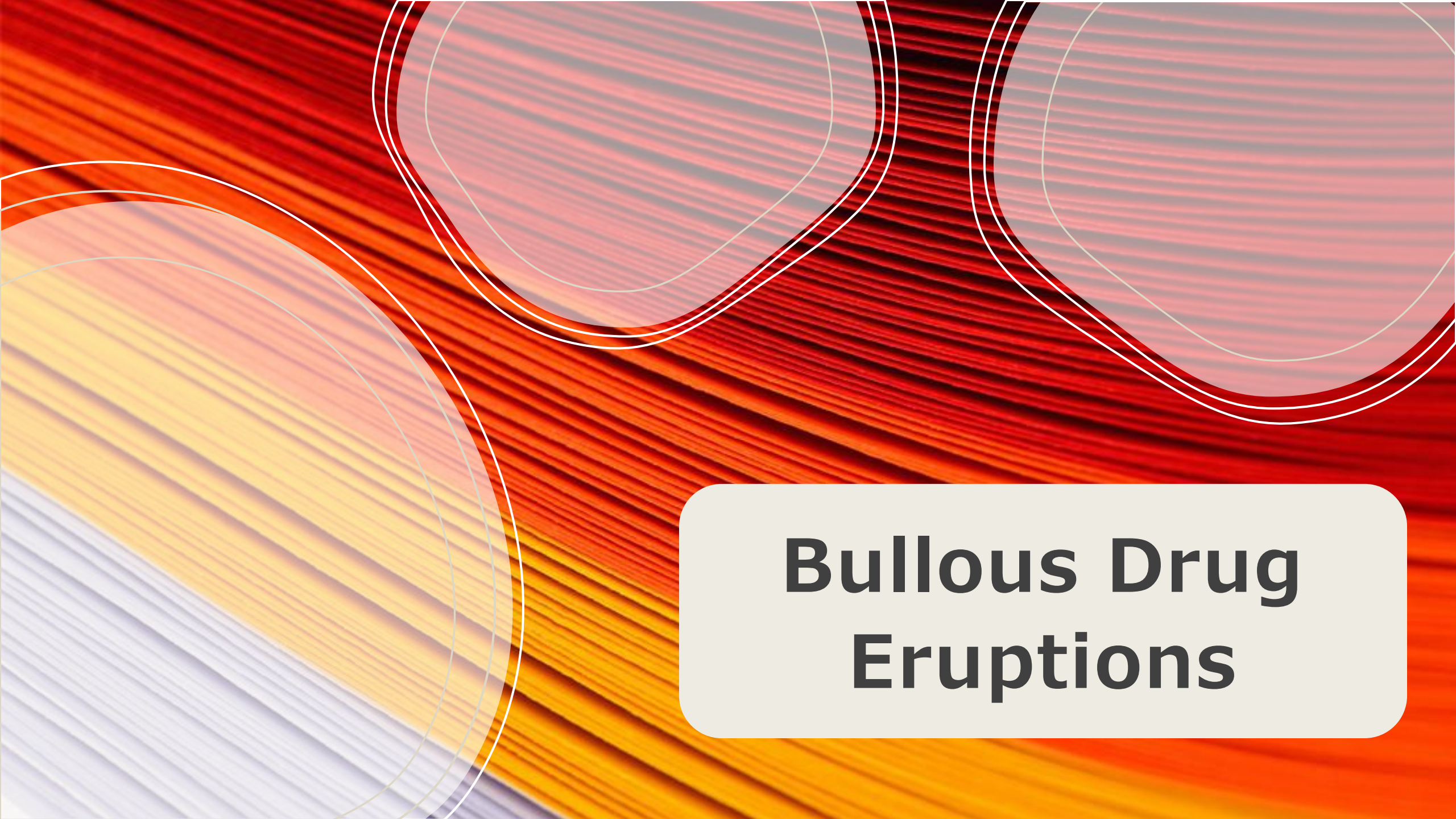


Herpes zoster



- Reactivation of varicella zoster virus
- Risk factors: immunocompromised state
- Dermatomal distribution of vesicles to disseminated
- Antiviral therapy (ex: valacyclovir)
- Pain management (acetaminophen, gabapentin, steroids)





Bullous Drug Eruptions

Bullous Drug Eruptions

- Unlike the autoimmune blistering diseases—which typically have a specific cleavage plane in skin—bullous drug eruptions involve *all layers of epidermis*
- Pathogenesis: scattered or complete keratinocyte necrosis in all layers of epidermis
- We'll be talking about drug eruptions
 - Erythema multiforme
 - Stevens-Johnson syndrome
 - Toxic epidermal necrolysis
- Tx for all
 - **Stop the causative medication**

Erythema Multiforme

- Etiology
 - HSV, mycoplasma pneumoniae, covid-19
 - Medications (ex: penicillins)
- Clinical presentation
 - Target lesions.
 - Can be atypical (raised, edematous palpable lesions with only 2 zones of color).
 - Can have mucosal involvement
- Tx: stop causative med, tx infectious cause
→ depending on severity, topical or systemic steroids



Stevens-Johnson Syndrome and Toxic Epidermal Necrolysis

- Continuum of **severe** cutaneous extensive necrosis and detachment of epidermis. Based on % of skin body surface area (BSA) detached
 - **<10% of BSA detached – SJS**
 - 10-30% of BSA detached – SJS/TEN overlap
 - **>30% of BSA detached – TEN**
- Mucous membranes are affected
- Cause:
 - in most cases, **due to medication (rxn starts 1 wk to 1month—sometimes up to 2 months—after starting medication)**
 - **Anticonvulsants (lamotrigine, carbamazepine, phenytoin)**
 - **Allopurinol**
 - **NSAIDS (-oxicam)**
 - **Antibiotics**
 - Mycoplasma infection
 - idiopathic

Stevens-Johnson Syndrome and Toxic Epidermal Necrolysis



- Clinical presentation:

- **Prodromal symptoms** (fever, malaise, myalgia, sore throat, conjunctivitis)
- **Exanthematous rash** → start on face and thorax, then spread to other areas symmetrically → lesions coalesce → over 1-2 days, will develop dusky erythema, purpuric spots, atypical targets, and flaccid bullae → bullae erupt, and get sheetlike detachment → erosions
- **+ Nikolsky sign**



Stevens-Johnson Syndrome and Toxic Epidermal Necrolysis

- Complications
 - Fluid and electrolyte imbalance
 - Sepsis
 - Hypothermia
 - Other organ involvement
 - death

Stevens-Johnson Syndrome and Toxic Epidermal Necrolysis

- Treatment:
 - *Immediate hospitalization, burn center*
 - *Stop causative drug*
 - SCORTEN score for prognosis
 - Supportive care (fluid and electrolyte management, wound care, nutrition, temperature, pain control, infection prevention)
- Prognosis:
 - 25% mortality from sepsis, acute respiratory distress syndrome, disseminated intravascular coagulation, multi-organ failure



Other

Edema Bullae

- Occurs in areas of dependent edema
- Arms or legs
- Setting of acute, severe venous stasis or anasarca
- Ex: CHF, renal failure, liver cirrhosis



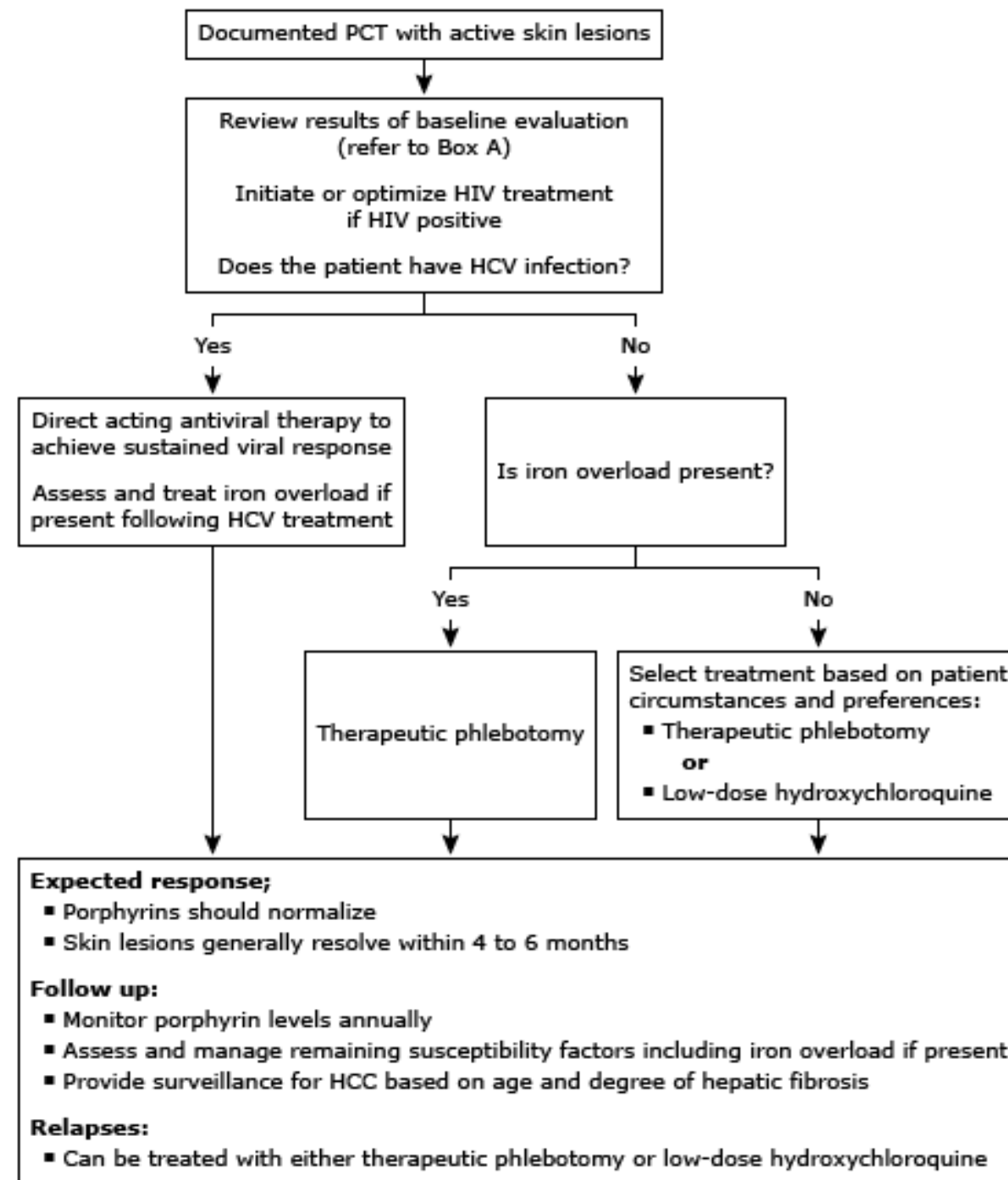
Porphyria Cutanea Tarda

- Most common porphyria
- Caused by **lack of uroporphyrinogen decarboxylase**
- Triggers: **alcohol**, estrogens, dialysis, viral infections (hep C, HIV)
- Associated with **hemochromatosis**
- Uroporphyrins cause blistering in skin with exposure to blue light
- Clinical presentation: **bullae in sun-exposed distribution**, milia of the dorsal hand, hypertrichosis of face, photosensitivity. May also see scarring alopecia with sclerodermoid changes in skin in advanced cases



Porphyria Cutanea Tarda

- Diagnosis:
elevated uroporphyrins in serum and 24-hour urine
- Treatment:
 - Sun protection



Box A

Baseline evaluation:

- Susceptibility factors
 - Alcohol
 - Smoking
 - Estrogen use
- Laboratory
 - CBC
 - Ferritin
 - UROD genetic testing
 - HFE genetic testing
 - LFTs and creatinine
 - HCV testing
 - HIV testing
- Liver imaging for liver disease
- HCC surveillance for advanced liver fibrosis or cirrhosis
- Baseline retinal examination if hydroxychloroquine is planned

Pseudoporphyria

- Resembles Porphyria Cutanea Tarda, but no abnormality in porphyrin metabolism
- Associated with **chronic renal insufficiency**
- Triggers
 - **Drugs** (NSAIDS, retinoids, furosemide, antibiotics)
 - **Tanning bed use**
- Treatment
 - Avoid triggers

Allergic Contact dermatitis

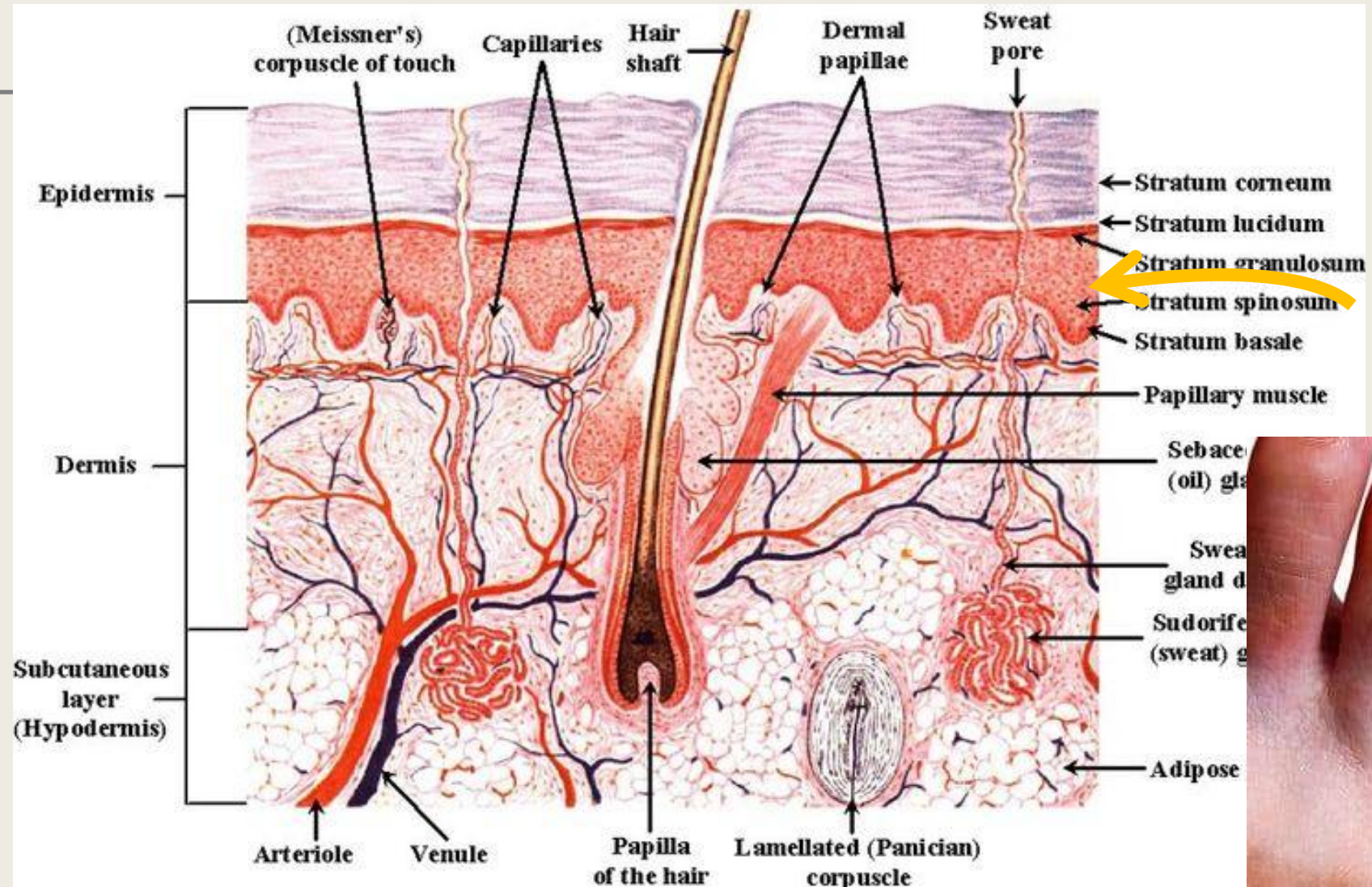
- Especially to **poison IVY**
- Extreme inflammation → rapid accumulation of intraepidermal interstitial fluid → **blistering**
- Treatment: **topical or systemic steroids, trigger avoidance**



Friction Blisters



- Typically occur on the feet
- Blistering plane is the **spinous layer**



Miliaria



- Results from **clogged sweat ducts**
- Depending on level of sweat duct occlusion:
 - Miliaria crystallina → stratum corneum
 - Miliaria rubra → spinus layer
 - Miliaria profunda → dermoepidermal junction
- 1st 2 can present with **tiny vesicles at site of occlusion**
 - Ex: on back of a warm, bed-ridden patient
- Treatment
 - **Minimize heat and humidity (go to cooler environment, breathable cotton clothing)**
 - Mild topical steroids (hydrocortisone) and topical abx if needed (clindamycin)

Osteopathic Considerations

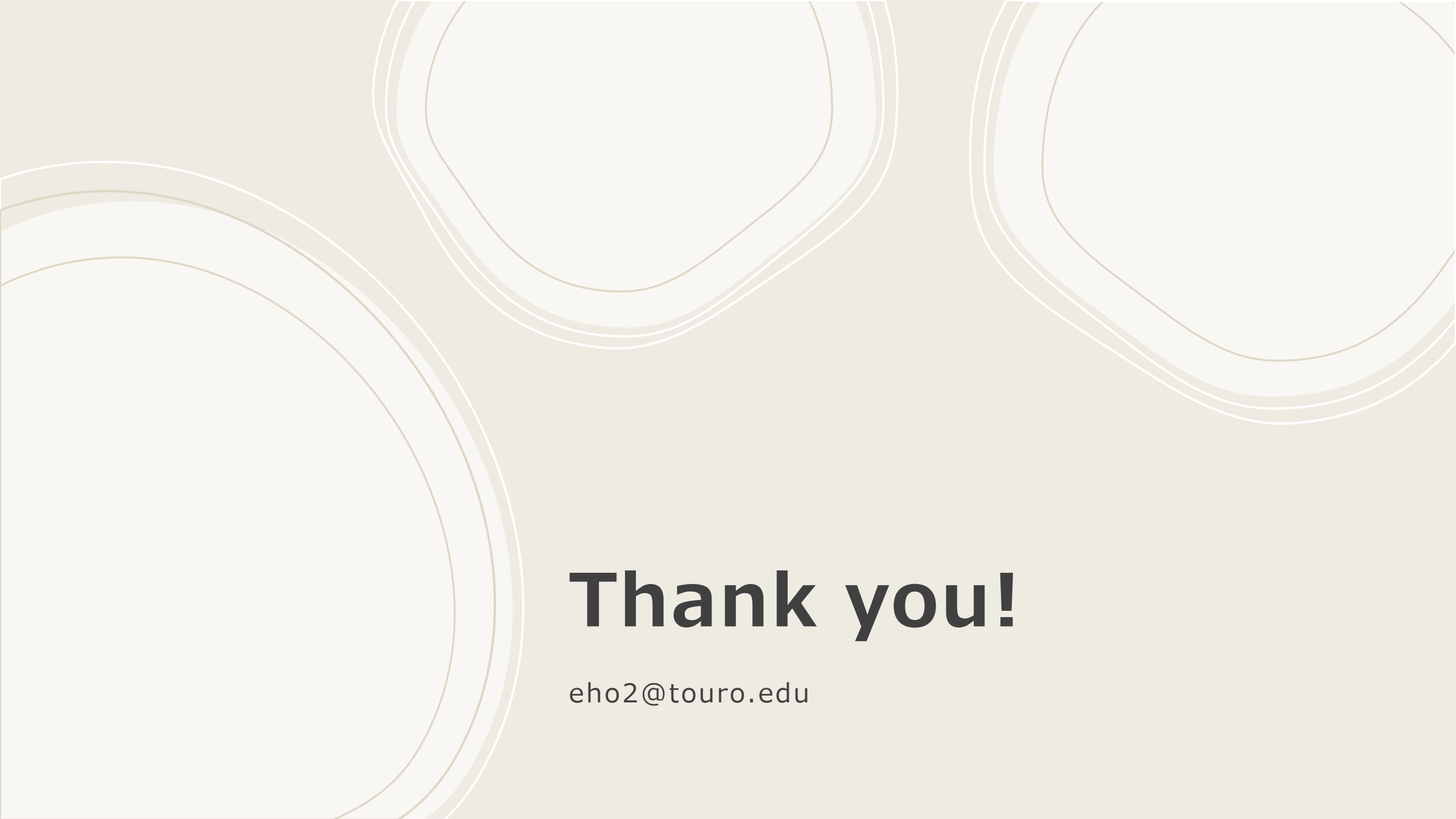
◆ Autonomics – optimize parasympathetics with treatment of OA and sacrum. Rib raising to balance sympathetics and help with lymphatic flow

◆ Biomechanics – treat somatic dysfunction near site of lesions. Indirect and gentle techniques to minimize friction.

◆ Circulation – myofascial release to diaphragms (thoracic inlet, abdomen, pelvic) to help lymphatic flow

◆ Screening – Zink screen, diet, movement/exercise, behavioral health support, social support, caregiver support

◆ Be mindful with open wounds, assist with pain management via OMT



Thank you!

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